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Human CD1c-autoreactive T cells recognise *Mycobacterium tuberculosis*-infected antigen-presenting cells and display cytotoxic effector programmes

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eLife Assessment

The study investigates how CD1c-restricted T cells respond to Mtb-infected APCs, leading to increased cytokine production and cytotoxic activity that may help control Mtb infection. While the work is **important** and will interest researchers in the field, the supporting evidence is **incomplete** and could be strengthened by additional experiments. Experiments would: (i) evaluate THP1-CD1c cells to determine whether MHC surface expression is reduced or entirely abolished, (ii) enhance confidence in the purity of the CD1c-specific T cell population isolated from blood, and (iii) suggest what additional signal THP1-CD1c cells treated with Mtb express that is absent from the untreated cells.

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Abstract

Tuberculosis (TB), caused by *Mycobacterium tuberculosis* (Mtb), remains the leading cause of death from infection globally, yet the contribution of non-classical T-cell pathways to human immunity remains poorly defined. CD1c-autoreactive T-cells, which recognise self-lipids presented by the antigen-presenting molecule CD1c, are frequent in human blood, but their role during infection is unclear. Here, we investigate how CD1c-expressing antigen-presenting cells (APCs) and Mtb infection shape CD1c-autoreactive T-cell responses using engineered human APC systems, complemented by single-cell transcriptomic profiling to define the *ex vivo* phenotypic landscape of these T-cells. CD1c is present within human TB granulomas, whereas Mtb down-modulates CD1c expression on infected APCs, consistent with an immune evasion strategy. CD1c-autoreactive T-cells respond more strongly to Mtb-infected CD1c+ APCs than to uninfected cells, exhibiting enhanced activation, cytotoxicity, and diverse cytokine secretion via CD1c-dependent recognition. Under *in vitro* conditions, these T-cells reduce relative Mtb burden in infected phagocytes. Single-cell RNA-sequencing reveals cytotoxic effector-memory programmes and expression of antimicrobial molecules, providing a mechanistic basis for these responses. Together, these findings define a human CD1c-restricted T-cell response to Mtb-infected APCs and identify autoreactive CD1c-restricted T-cells as a candidate cellular axis for lipid-directed immunity in TB.

Introduction

Tuberculosis (TB), caused by *Mycobacterium tuberculosis* (Mtb), is a major global epidemic ^{1–3}. The only licensed vaccine, Bacille Calmette–Guérin (BCG), has variable efficacy against adult pulmonary disease, which drives transmission, and therefore new vaccines are urgently needed ⁴. Typically, novel TB vaccines have focussed on eliciting conventional T-cell responses to peptide antigens presented by highly polymorphic major histocompatibility complex (MHC) molecules ^{5,6}. Although progress has been made in developing new immunisation tools, one of the most promising candidates, M72, based on immunogenic peptides, achieved only 50% efficacy against TB ^{7–10}. This partial efficacy highlights the need to explore alternative approaches, including non-peptide antigens, to achieve the level of protection required for global TB control. Consequently, there is growing interest in harnessing unconventional T-cell responses as a complementary or alternative strategy ^{11,12}. Generally, unconventional T-cells recognise antigens bound to non-polymorphic antigen presenting molecules and therefore are not restricted by genetic variability. This group includes MR1-restricted mucosal associated invariant T-cells (MAIT), CD1d-restricted invariant natural killer T-cells (iNKT), HLA-E- and CD1 group 1-restricted T-cells ^{11,13}.

The CD1 family (CD1a, CD1b, CD1c, CD1d) are non-polymorphic MHC-class I-like proteins that bind and present self or foreign lipid antigens to $\alpha\beta$ and $\gamma\delta$ T-cells ¹⁴. CD1c is the most ubiquitously expressed group 1 CD1 molecule, present on antigen presenting cells (APCs) such as dendritic cells (DCs), foamy macrophages and B-cells ^{15,16}. Several studies suggest a role for CD1c in host immunity to Mtb. CD1c presents Mtb-derived lipids to T-cells, including mannosyl- β 1-phosphomycoketide (MPM) and phosphomycoketide (PM) ^{17–21}. CD1c-restricted and Mtb lipid-specific T-cells expand in the circulation of TB patients, and CD1c-PM loaded tetramers detect PM-specific T-cells in individuals with latent TB ^{18,22}.

However, it is notable that the majority of CD1c-restricted T-cells exhibit autoreactivity against CD1c expressing APCs in the absence of exogenous microbial antigens, suggesting recognition of CD1c bound to endogenous self-lipids ^{23,24}. These CD1c-autoreactive T-cells recognise diverse self-lipids including phospholipids, cholesteryl-esters, methyl-lysophosphatidic acids (mLPAs), and monoacylglycerols (MAG) ^{25–27}. CD1c-autoreactive T-cells are relatively abundant in the circulation of healthy adults and are proposed to become activated in response to host lipids in autoimmune disease and cancer ²³. CD1c-autoreactive T-cells isolated from patients with systemic lupus erythematosus (SLE) provide help to CD1c⁺ B-cells, enhancing the production of IgG and contributing to immunopathology ²⁸. CD1c-autoreactive T-cells infiltrate lesions in Hashimoto's thyroiditis and Graves' disease, contributing to tissue destruction ²⁹. CD1c-autoreactive T-cells recognise tumour associated self-lipid antigens (mLPA) derived from leukemic cells ²⁷. mLPA-specific T-cells efficiently kill CD1c⁺ acute leukaemia cells and protect immunodeficient mice against CD1c⁺ human leukaemia cells ²⁷. Therefore, CD1c-autoreactive T-cells contribute to the immune response in human autoimmune disease and tumour immunosurveillance. However, historically these diseases will not have exerted the dominant evolutionary pressure relative to mortality from infection.

Humans have co-evolved with bacteria including gut commensals ³⁰ and environmental and pathogenic mycobacteria ³¹. This long-standing relationship with microorganisms may have exerted evolutionary pressure on CD1c-reactive T-cells to contribute to immune defence against pathogenic organisms ³¹. Nevertheless, the extent to which CD1c-autoreactive T-cells participate in human responses to infection has not been defined. Indeed, CD1c-autoreactive T-cells expand after Mtb infection of CD1 transgenic mice (hCD1Tg) *in vivo* ³². Furthermore, autoreactive T-cells restricted by other CD1 family members also exhibit enhanced responses to TLR agonists, mycobacteria, and mycobacterial lipids, thereby indicating that these T-cells may also be activated during infection ^{33,34}. These observations suggest a potential, but as yet uncharacterised, role for CD1c-autoreactive T-cells during microbial infection. Consequently, we hypothesised that CD1c-autoreactive T-cells may play a role in anti-bacterial protection. Here, we sought to define how CD1c-autoreactive T-cells respond to Mtb-infected antigen-presenting cells using human *in vitro* systems. We first confirm their inherent autoreactivity, as they activate and release cytokines in

response to CD1c⁺ APCs in the absence of exogenous lipid antigens. We then demonstrate CD1c-autoreactive T-cells display greater activation, cytotoxicity and cytokine secretion in response to Mtb-infected CD1c⁺ APCs compared to their baseline autoreactive response. The enhanced reactivity is dependent on TCR recognition of CD1c on infected APCs. CD1c-autoreactive T-cells exhibit anti-microbial activity, reducing Mtb growth in phagocytic cells. Finally, single-cell transcriptomic profiling of *ex vivo* CD1c-autoreactive T-cells revealed a predominant cytotoxic effector memory phenotype and expression of antimicrobial molecules, providing mechanistic insight into effector functions. Together, these data identify a human autoreactive CD1c-restricted T-cell response to Mtb-infected APCs and demonstrate that autoreactive CD1c-restricted T cells contribute to lipid-mediated immunity during TB infection.

Results

Circulating CD1c-autoreactive T-cells in healthy donors

CD1c is an MHC class I-like molecule consisting of a heavy chain and β 2-microglobulin (β 2m) light chain²⁶. CD1c-autoreactive T-cells are defined by their recognition of CD1c in the absence of exogenous lipid antigen^{24,25,27,35}. To investigate the presence of CD1c-autoreactive T-cells in the blood of healthy donors, we developed novel *in vitro* assays to allow their specific expansion and subsequent detection by flow cytometry. We generated β 2m knock-out THP1 monocytic leukaemia cells, THP1-KO, that lack the expression of β 2m and therefore do not express MHC-class I and CD1 proteins on the cell surface. To provide specific CD1c stimulation, β 2m knock-out THP1 monocytic leukaemia cells were then transduced with a CD1c- β 2m fusion gene construct, THP1-CD1c, which express high levels of CD1c and do not express MHC-class I (Fig. 1A [↗](#)).

Purified CD3⁺ T-cells isolated from 14 healthy donors were labelled with the proliferation marker CellTrace violet (CTV) and co-cultured with irradiated THP1-CD1c APCs without exogenously added lipid antigens. After 12 days *in vitro* culture, a fraction of cells had undergone cell division, as indicated by dilution of CTV, consistent with the presence of autoreactive T-cells (Fig. S1A [↗](#)). To confirm this, we performed an activation-induced marker (AIM) assay and observed that a significant fraction of proliferating cells expressed the activation-induced markers CD69, CD25 and CD137 upon re-challenge with THP1-CD1c cells (Fig. 1B [↗](#) and 1C [↗](#)), with donor-to-donor variation in expansion. Importantly, CD1c-autoreactive T-cells expanded in most donors, with increased expression of CD69, CD25 and CD137 after overnight stimulation with THP1-CD1c cells (Fig. 1C [↗](#)). We performed a Luminex assay on short term cultures expanded by THP1-CD1c APCs to measure cytokine secretion, studying proinflammatory and anti-inflammatory cytokines known to be critical in anti-mycobacterial immunity⁵. CD1c-mediated stimulation, without exogenously added antigen, induced production of several cytokines including IL-1 β , IL-2, IL-10, IL-13 and GM-CSF, suggesting a polyfunctional phenotype (Fig. S1B [↗](#)). Thus, CD1c-autoreactive T-cells can be detected in healthy donors and expanded in short-term *in vitro* culture.

CD1c expression in human TB granulomas

We hypothesised that CD1c-autoreactive T-cells may have a role in mycobacterial infection, and so first investigated CD1c expression at the site of TB infection, which has not been previously characterised. We performed immunohistochemical staining of lung biopsies taken from patients with active pulmonary TB with a recombinant anti-CD1c antibody. Expression of CD1c was observed in lung tissue from all five TB patients (Fig. 2A [↗](#)). Areas of positive staining were primarily observed at distal inflammatory tissue remote from the granuloma centre and in B-cell follicles (Fig. 2Av [↗](#) and 2Avi). CD1c staining was observed within some granulomas (Fig. 2Ai-iii [↗](#)), however overall immunoreactivity within central granuloma areas was not frequently observed. Therefore, CD1c expression within infected lung tissue was confirmed, but predominantly remote from the centre of TB granulomas.

The low level of CD1c expression in the central granuloma led us to investigate the dynamics of CD1c expression during Mtb infection. Previous reports suggested that Mtb regulates the antigen presenting functions of DCs by downregulating CD1c expression, thereby repressing their

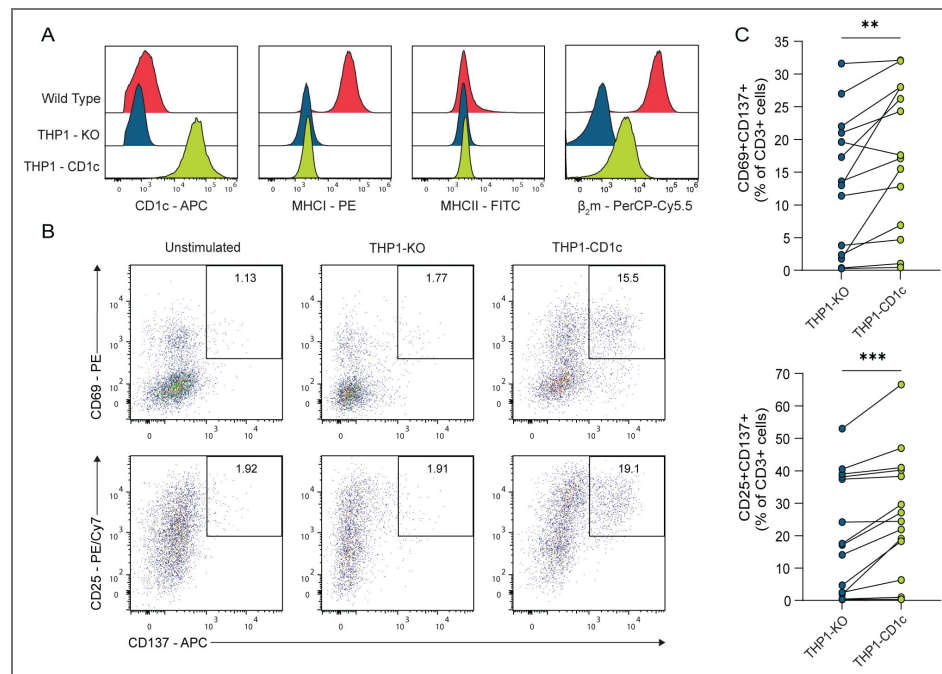


Figure 1. Autoreactive T-cells expand upon CD1c stimulation.

(A) Representative histogram overlays showing CD1c, MHCI, MHCII, and β_2m expression on wild type THP1 cells, and engineered THP1-KO and THP1-CD1c APCs. (B) Representative flow cytometry analysis of T-cells from a healthy donor expanded with THP1-CD1c cells in the absence of exogenous lipid antigen and then cultured overnight with THP1 cells. Cells were gated on live CD3⁺CTV⁻ T-cells and T-cell activation was measured with anti-CD69, anti-CD25 and anti-CD137. Plots show expression of CD69 and CD137 (top), CD25 and CD137 (bottom). Significant numbers of CD1c-autoreactive T-cells were present in expanded cultures. (C) Cumulative data from 14 healthy donors showing frequency of CD3⁺CTV⁻CD69⁺CD137⁺ T-cells (top), and CD3⁺CTV⁻CD25⁺CD137⁺ T-cells (bottom) from lines first expanded with THP1-CD1c and then overnight culture with THP1-KO or THP1-CD1c APCs. CD1c-autoreactive T-cells were present in the majority of donors. ** $P < 0.01$; *** $P < 0.001$; (C, Wilcoxon matched pairs signed rank test).

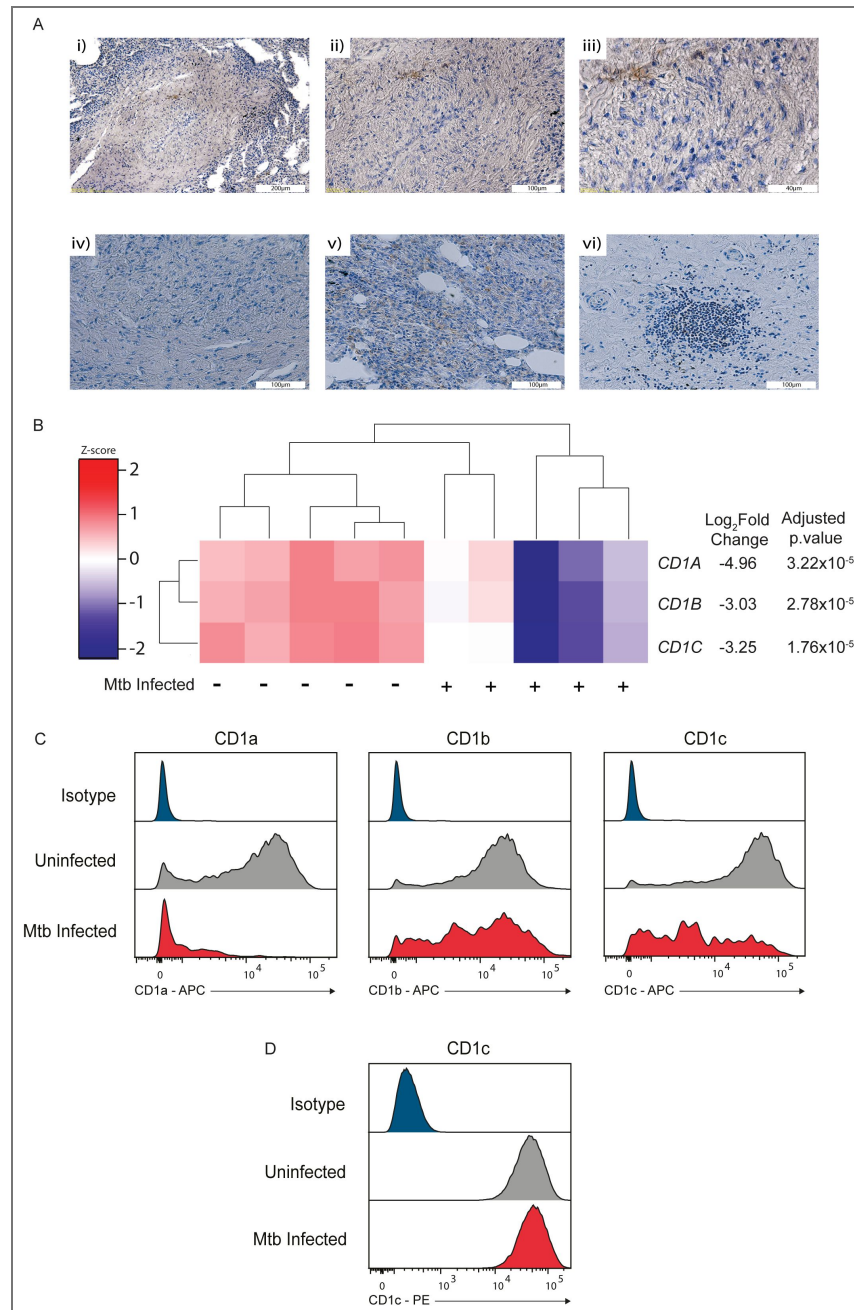


Figure 2. CD1c is expressed in human TB and downregulated in APCs by Mtb infection.

(A) Lung biopsies from patients with active TB were stained with anti-CD1c antibodies. i-iii) Anti-CD1c immunoreactivity in a representative TB granuloma. iv) Control staining of the same granuloma with secondary antibody only and avidin biotin-peroxidase complex (ABC) detection shows no immunoreactivity. v) Extensive staining in inflammatory tissue remote from the TB granuloma. vi) CD1c staining in B-cell follicles adjacent to the granuloma. (B) RNA-Seq heat map showing significant reduction in CD1A, CD1B, and CD1C expression in MoDCs from five donors at 48 hours after Mtb infection (MOI=1). Differential gene expression was performed on filtered normalised counts using the voom – limma pipeline in R Studio with p-value adjustment being performed using the Benjamini-Hochberg method. Differentially expressed genes were identified as having Log₂FC > +1 (Upregulated) or < -1 (Downregulated) with adjusted p-values < 0.05. (C) MoDC histograms of normalised counts demonstrating a reduction in CD1a, CD1b, and CD1c expression on differentiated MoDCs following live Mtb infection (MOI=1). Data representative of an experiment conducted in two donors performed in triplicate. (D) Flow cytometry histograms showing CD1c expression on THP1-CD1c cells at 72 hours following infection with live Mtb (MOI=1). Mtb infection does not change CD1c expression on THP1-CD1c cells. Data representative of two experiments performed in triplicate.

response to infection ^{36,37}. To investigate the impact of Mtb infection on CD1c expression, we first interrogated an RNA-sequencing dataset derived from five healthy donors where monocyte derived DC (MoDC) were infected with live Mtb ³⁸. At 48 hours after Mtb infection, CD1 group 1 expression was significantly reduced in infected MoDCs derived from all five donors relative to uninfected cells (Fig. 2B [↗](#)). Next, we validated these results at the protein level on MoDCs derived from healthy donors by flow cytometry. Our results corroborate the transcriptomic dataset and reveal significantly reduced expression of CD1c protein on primary MoDCs after 48 hours of live Mtb infection measured by flow cytometry (Fig. 2C [↗](#) & Fig. S2A [↗](#)). To investigate whether Mtb regulates CD1c expression on our engineered THP1-CD1c APCs, we measured CD1c expression after Mtb infection. Contrary to the reduced expression of CD1c on infected primary MoDCs, CD1c expression on THP1-CD1c APCs did not change following Mtb infection (Fig. 2D [↗](#) & Fig. S2B [↗](#)). Overall, our results reveal CD1c expression in Mtb-infected lung tissue, but primarily away from the granuloma centre, and Mtb down-regulates CD1c on DCs. Importantly, Mtb infection does not suppress CD1c expression on the engineered THP1-CD1c cells, supporting the use of this APC model to study CD1c-mediated T-cell responses.

CD1c-autoreactive T-cells are cytotoxic

To investigate CD1c-autoreactive T-cell function in more detail, we attempted to derive pure T-cell lines from the peripheral blood of healthy donors. We utilised two different approaches to achieve this. Firstly, we used the THP1-CD1c APC system to expand T-cells from a donor that exhibited good T-cell expansion (Fig. 1B [↗](#)). In a parallel approach, we employed CD1c-endo streptamers to enrich blood derived T-cells from another donor (Fig. S3). After flow cytometry sorting and expansion of T-cells in culture, polyclonal T-cell lines from two donors were derived that stained brightly with CD1c-endo tetramers (Fig. 3A [↗](#) and 3C [↗](#)). Both T-cell lines expressed $\alpha\beta$ TCRs and CD4 co-receptors (Fig. 3B [↗](#) and 3D [↗](#)). Importantly, T-cell functional assays revealed activation of T-cells in response to THP1-CD1c but not parental THP1-KO APCs, as shown by their upregulation of the T-cell activation markers CD69 and CD25 (Fig. 3E [↗](#)).

CD1c-autoreactive T-cells have been reported as having cytotoxic activity ^{27,39}, and so we next investigated their cytotoxicity against target cells stimulated with UV-killed TB. We first cultured CD1c-autoreactive T-cells with THP1-CD1c APCs treated with increasing doses of UV-killed Mtb and measured cell death. CD1c-autoreactive T-cells mediated cytotoxicity toward THP1-CD1c APCs treated with UV-killed Mtb, in a Mtb dose dependent manner, while no cytotoxicity occurred against untreated THP1-CD1c APCs (Fig. 3F [↗](#)). Irrelevant control T-cells did not induce notable cytotoxicity against THP1-CD1c APCs (Fig. 3F [↗](#)). In addition, CD1c-autoreactive T-cells induced significant cytotoxicity, in a T-cell dose dependent manner, against THP1-CD1c APCs treated with UV-killed Mtb (MOI=10), but not to untreated controls (Fig. 3G [↗](#)). Increasing the dose of the control T-cells did not induce cytotoxicity (Fig. 3G [↗](#)). These results demonstrate that CD1c-autoreactive T-cells are cytotoxic towards CD1c⁺ APCs treated with UV-killed Mtb.

CD1c-autoreactive T-cells exhibit enhanced lysis of Mtb-infected APCs beyond baseline autoreactivity

To study T-cell cytotoxicity in more detail, we performed complimentary assays studying live Mtb infection. We cultured T-cells with THP1-KO and THP1-CD1c that were either uninfected or infected with live Mtb. First, we investigated T-cell activity by measuring the activation markers CD69 and CD25 by flow cytometry. CD1c-autoreactive T-cells were activated when cultured with uninfected THP1-CD1c APCs but not with control THP1-KO APCs, consistent with CD1c-dependent autoreactivity (Fig. 4A [↗](#) and 4B [↗](#)). Notably, increased activation of CD1c-autoreactive T-cells occurred when they were cultured with Mtb-infected THP1-CD1c APCs (Fig. 4A [↗](#) and 4B [↗](#)). To investigate T-cell cytotoxicity more directly, we cultured the T-cells with APCs before measuring target cell viability by flow cytometry (Fig. 4C [↗](#)). Importantly, we observed lysis of Mtb-infected THP1-CD1c APCs by CD1c-autoreactive T-cells, greater than the lysis of uninfected THP1-CD1c APCs (Fig. 4D [↗](#)).

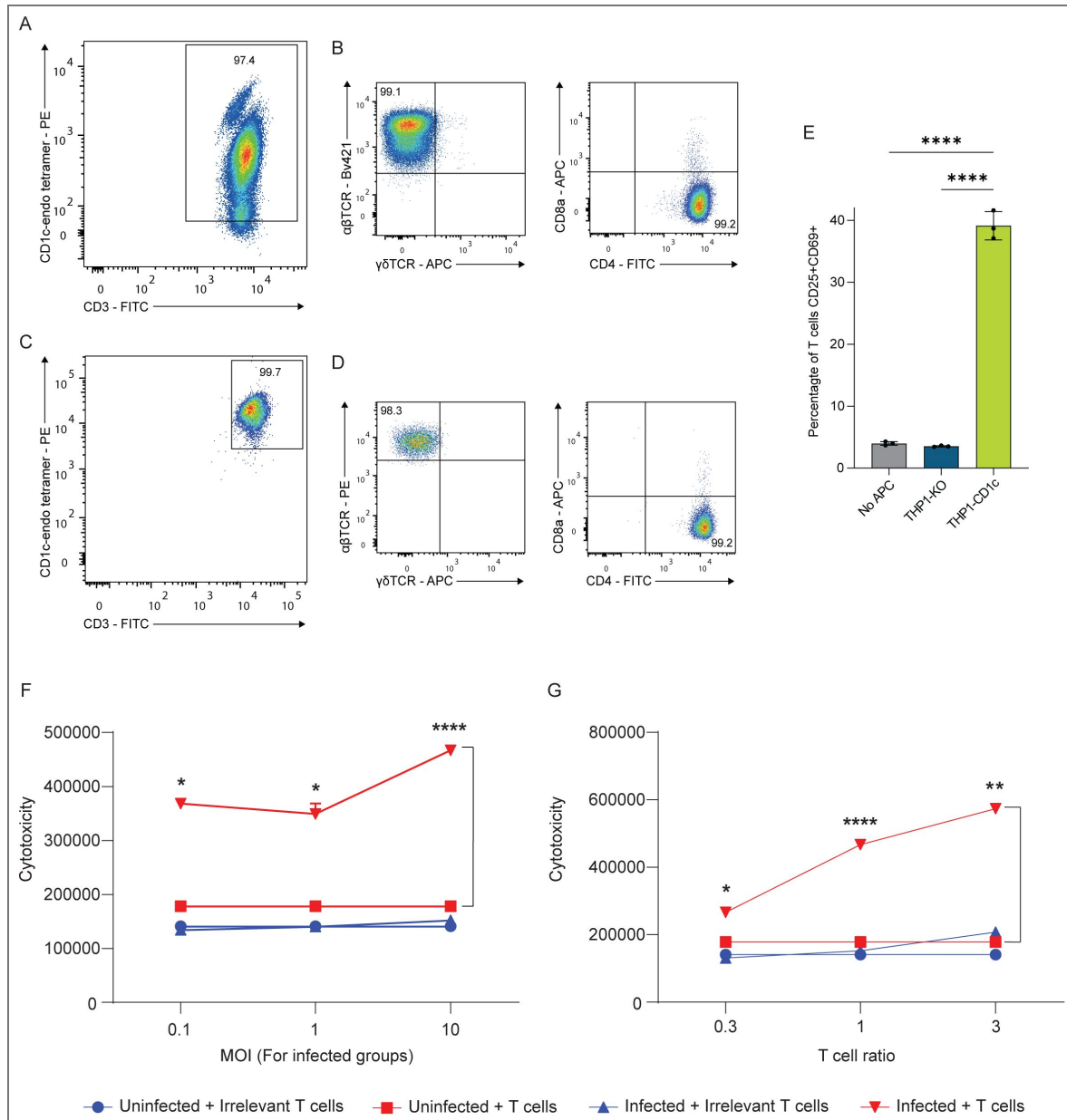


Figure 3. CD1c-autoreactive T-cells are cytotoxic to Mtb stimulated cells.

(A-D) CD1c-endo tetramer staining of T-cell lines generated from two healthy donors. (A) T-cells generated from PBMCs after CD1c-endo streptamer enrichment and one round of CD1c-endo dextramer flow cytometry sorting and subsequent *in vitro* expansion (Fig. S3). (C) T-cells generated after expansion with THP1-CD1c cells, followed by CD1c-endo-tetramer guided cell sorting and subsequent *in vitro* expansion. (B) and (D) FACS plots demonstrating that T-cells in (A) and (C), respectively, are $\alpha\beta$ TCR and CD4 positive. (E) T-cell lines are activated in response to THP1-CD1c APCs. Rested T-cells and T-cells cultured with THP1-KO APCs served as control. T-cell activity was determined by measuring the upregulation of CD69 and CD25 by flow cytometry. Data is representative of three experiments from the two donor-derived lines performed in triplicate. (F) CD1c-autoreactive T-cells display significant cytotoxicity in a Mtb dose dependent manner against UV-killed Mtb-treated THP1-CD1c APCs. No cytotoxicity was observed by T-cells cultured with untreated THP1-CD1c APCs or irrelevant control T-cells, lacking CD1c restriction. (G) CD1c-autoreactive T-cells displayed significant cytotoxicity in a T-cell dose-dependent manner against UV-killed Mtb-treated THP1-CD1c APCs. No cytotoxicity was observed by T-cells cultured with untreated THP1-CD1c APCs or with irrelevant control T-cells. Cytotoxicity was measured using a ToxiLight assay. Data is representative of two independent experiments, each preformed in triplicate. * $P < 0.05$; ** $P < 0.01$; **** $P < 0.0001$ (E, one-way ANOVA with Tukey's multiple comparison test; F-G, two-way ANOVA).

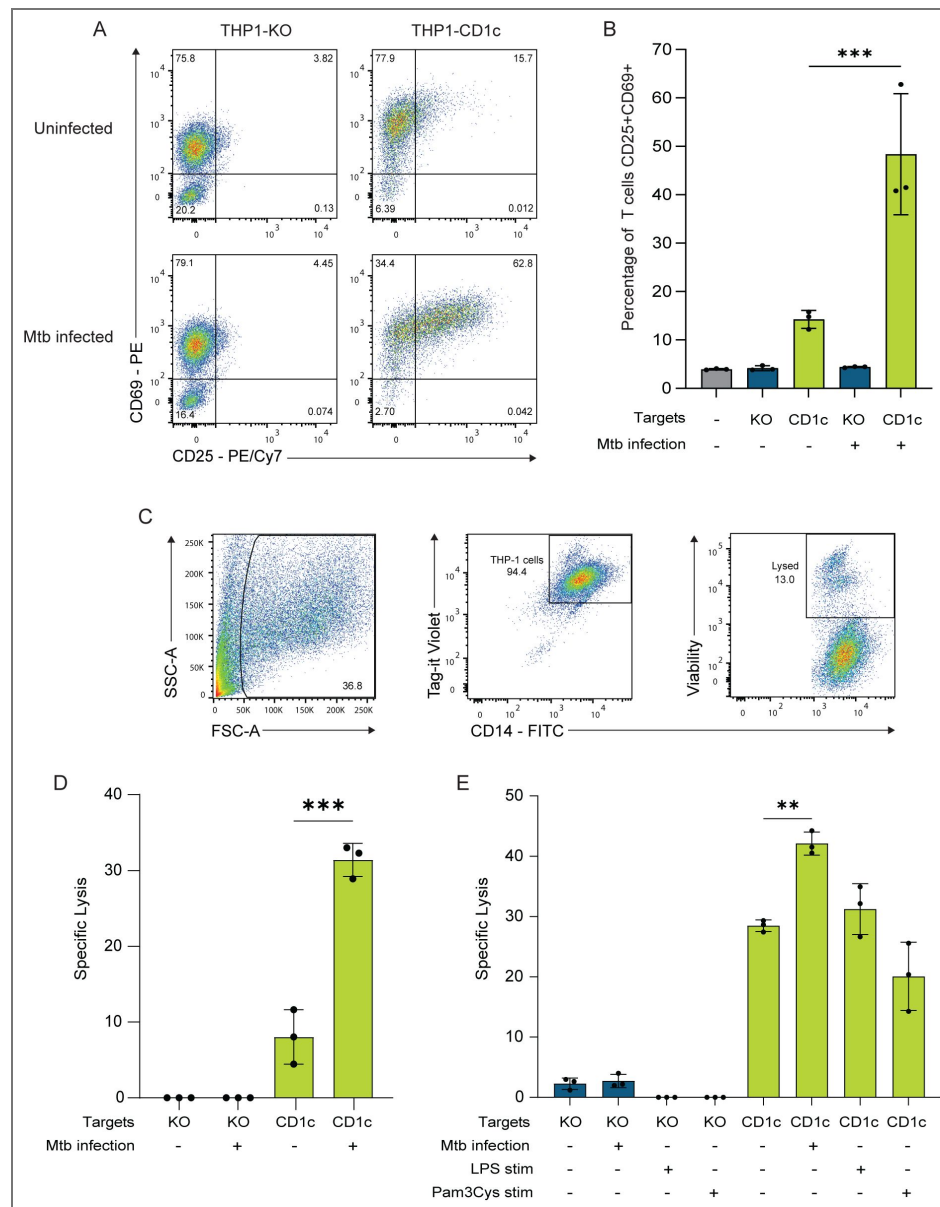


Figure 4. CD1c-autoreactive T-cells lyse target cells infected with live Mtb.

(A) Flow cytometry dot plots and (B) bar graphs showing CD1c-autoreactive T-cells are activated by THP1-CD1c APCs but not THP1-KO APCs, with significantly greater activation when THP1-CD1c APCs are infected with live Mtb (MOI=1). (C) Gating strategy for measuring the T-cell lysis of THP1 cells. Prior to T-cell culture, THP1 cells were stained with Tag-it violet to allow identification of THP1 target cells in co-culture. LIVE/DEAD Fixable Near-IR Dead Cell Marker was used to measure the proportion of lysed THP1 cells. (D) CD1c-autoreactive T-cells lyse THP1-CD1c APCs but not THP1-KO APCs. Killing is significantly enhanced when THP1-CD1c cells are infected with live Mtb (MOI=1). (E) Lysis of THP1-CD1c is increased with Mtb infection, but not enhanced with LPS or Pam3CSK4 stimulation. Data are representative of three independent experiments, each performed in triplicate. ** P < 0.01; ***P < 0.001 (B, D and E, one-way ANOVA with Tukey's multiple comparison test).

Recent studies suggest that activation of APCs with TLR agonists can enhance the activation of CD1c-autoreactive T-cells ^{33,40}. Therefore, to investigate whether the enhanced cytotoxicity of CD1c-autoreactive T-cells was facilitated via TLR mediated mechanisms, we cultured T-cells with THP1-CD1c APCs that were infected with Mtb or stimulated with the TLR agonists Pam3Cys (TLR2) or LPS (TLR4). TLR agonists did not enhance T-cell cytotoxicity above the autoreactive response, whereas Mtb infection of APCs induced significant cytotoxicity of THP1-CD1c APCs (Fig. 4E [↗](#)). Overall, our results confirm that CD1c-autoreactive T-cells kill Mtb-infected CD1c⁺ APCs, which is not replicated by TLR activation alone.

The enhanced Mtb reactivity is mediated via CD1c recognition by the TCR

To provide further mechanistic insight, we performed single-cell TCR sequencing of CD1c-autoreactive T-cells (Fig. 3A [↗](#)). TCR chain sequencing revealed the presence of two dominant TCR sequences in several wells. The EM1 TCR consists of an alpha chain containing the gene segments TRAV4 and TRAJ5 and a beta chain containing the gene segments TRBV7-8, TRBD1, and TRBJ2-7. The EM2 TCR consists of an alpha chain containing the gene segments TRAV1-2 and TRAJ26 and a beta chain containing the gene segments TRBV2, TRBD1, and TRBJ1-1 (Fig. 5A [↗](#)). We utilised the β 2m knock-out JRT3.5 Jurkat T-cell line, which lacks both TCR and CD1c expression, to avoid self-reactivity by Jurkat T-cells in functional studies (Fig. S4). We transduced the β 2m knock-out JRT3.5 Jurkat T-cell line with the EM1 and EM2 TCRs. Staining of TCR transduced Jurkats with CD1c-endo tetramers show clear binding of EM1 and EM2 Jurkat T-cells, consistent with direct engagement of CD1c by the TCR (Fig. 5B [↗](#)). CD1c-endo tetramers did not stain the control parental Jurkat T-cells (Fig. 5B [↗](#)). In addition, culturing the Jurkat T-cells on plates coated with CD1c-endo protein resulted in activation of both EM1 and EM2 Jurkat T-cell lines demonstrated by upregulation of CD69, and suggesting TCR-dependent signalling via CD1c engagement (Fig. 5C [↗](#)).

Next, we investigated EM1 and EM2 Jurkat T-cell activity against live Mtb infection using THP1 APCs as target cells. We observed Jurkat T-cell line activation through upregulation of CD69 when cultured with uninfected THP1-CD1c APCs but not with control THP1-KO APCs, confirming the autoreactivity of the TCRs (Fig. 5D [↗](#)). Importantly, we observed significantly more Jurkat T-cell activation when cells were cultured with Mtb-infected THP1-CD1c APCs (Fig. 5D [↗](#)). These data suggest that the enhanced activation of CD1c-autoreactive T-cells against Mtb-infected target cells is mediated by TCR recognition of CD1c.

CD1c-autoreactive T-cells release cytokines associated with TB immunity and suppress Mtb luminescence

To further investigate T-cell function in the context of Mtb infection, we cultured CD1c-autoreactive T-cells with APCs that were treated with UV-killed Mtb and analysed cytokine secretion. CD1c-autoreactive T-cell cytokine secretion was increased somewhat in response to untreated THP1-CD1c APCs, further corroborating their CD1c-autoreactivity (Fig. S5A [↗](#)). Critically though, CD1c-autoreactive T-cells secreted higher levels of cytokines in response to Mtb-treated THP1-CD1c APCs (Fig. 6A [↗](#) and 6B [↗](#)). No cytokines were released by an irrelevant control T-cell line (Fig. 6A [↗](#)), and the response was CD1c-mediated, with no upregulation in response to THP1-KO cells (Fig. S5B [↗](#)). To determine cytokines released by the APCs alone, we treated THP1-CD1c cells with Mtb and measured their cytokine release. Mtb-treated THP1-CD1c APCs released significant amounts of IL-8 and RANTES, but not any other cytokine (Fig. S6).

CD1c-autoreactive T-cells secreted the proinflammatory cytokines IFN- γ , TNF- α , IL-1 α , IL-2 and GM-CSF in response to Mtb-treated APCs, in a dose-responsive effect (Fig. 6A [↗](#) and B). Additionally, CD1c-autoreactive T-cells produced IL-4, IL-5, IL-10 and IL-13, demonstrating polyfunctionality. Finally, we assessed whether CD1c-autoreactive T-cells could restrict Mtb growth in phagocytes using luminescent live Mtb. Relative luminescence values for each condition were normalised to the matched infected APC-only control within each experimental replicate. These T-cells marginally reduced Mtb growth in THP1-KO APCs, likely due to basal cytokine secretion ⁴¹.

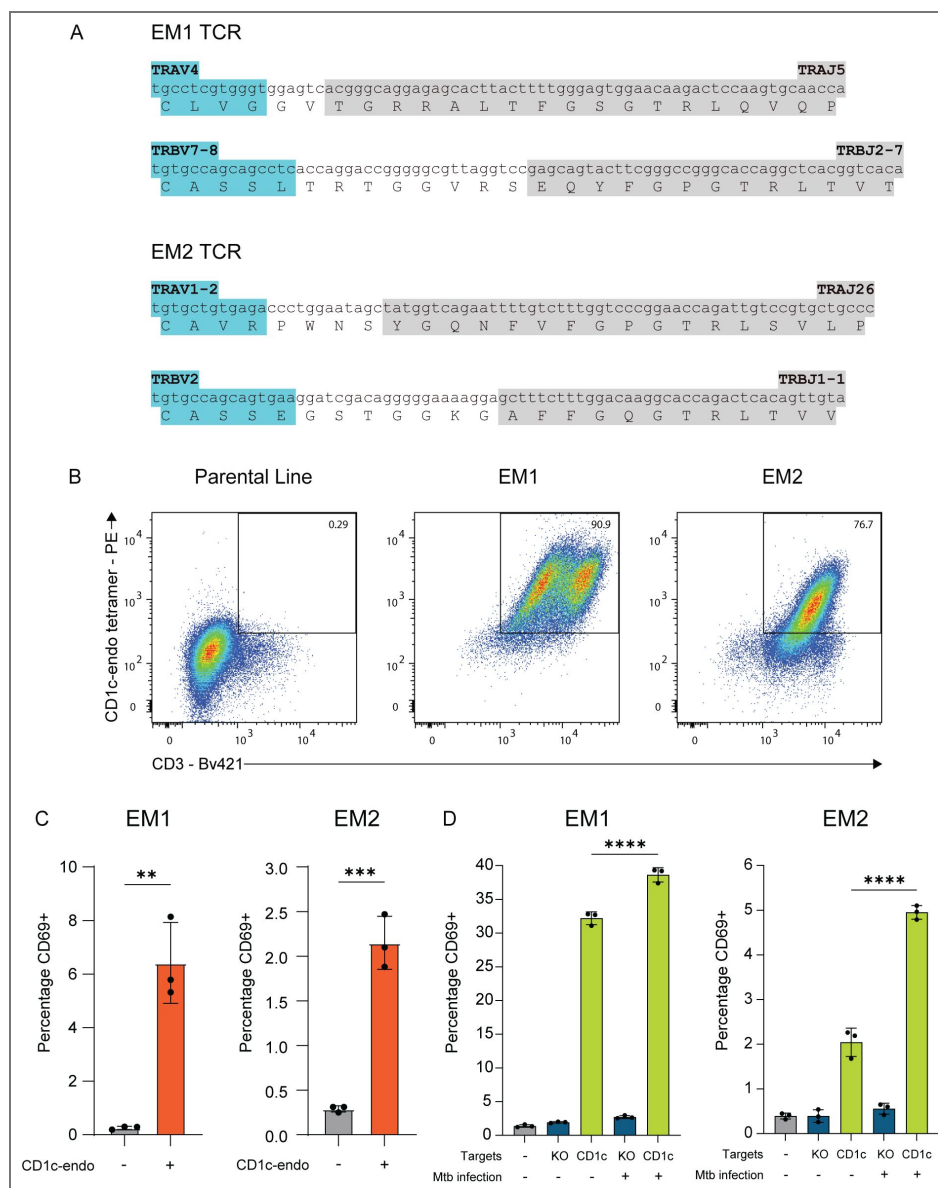


Figure 5. TCR-CD1c interactions mediate the response to Mtb infection.

(A) The alpha (top) and the beta (bottom) chain sequences of EM1 and EM2 TCRs. Variable region (grey), N additions (white) and joining segment (blue) are shown. (B) CD1c-endo tetramer staining of parental Jurkats and Jurkat T-cells transduced to express EM1 and EM2 TCRs. (C) Percentage activation of Jurkat T-cells transduced with EM1 and EM2 TCRs in response to uncoated or CD1c-endo coated wells. T-cell activity was measured by flow cytometry staining with anti-CD69. (D) Jurkat T-cells transduced with EM1 and EM2 TCRs are activated by THP1-CD1c APCs but not THP1-KO APCs, with significantly greater activation when THP1-CD1c APCs are infected with live Mtb (MOI=1). Data are representative of two independent experiments, each preformed in triplicate. ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$ (C, unpaired t-test, D, one-way ANOVA).

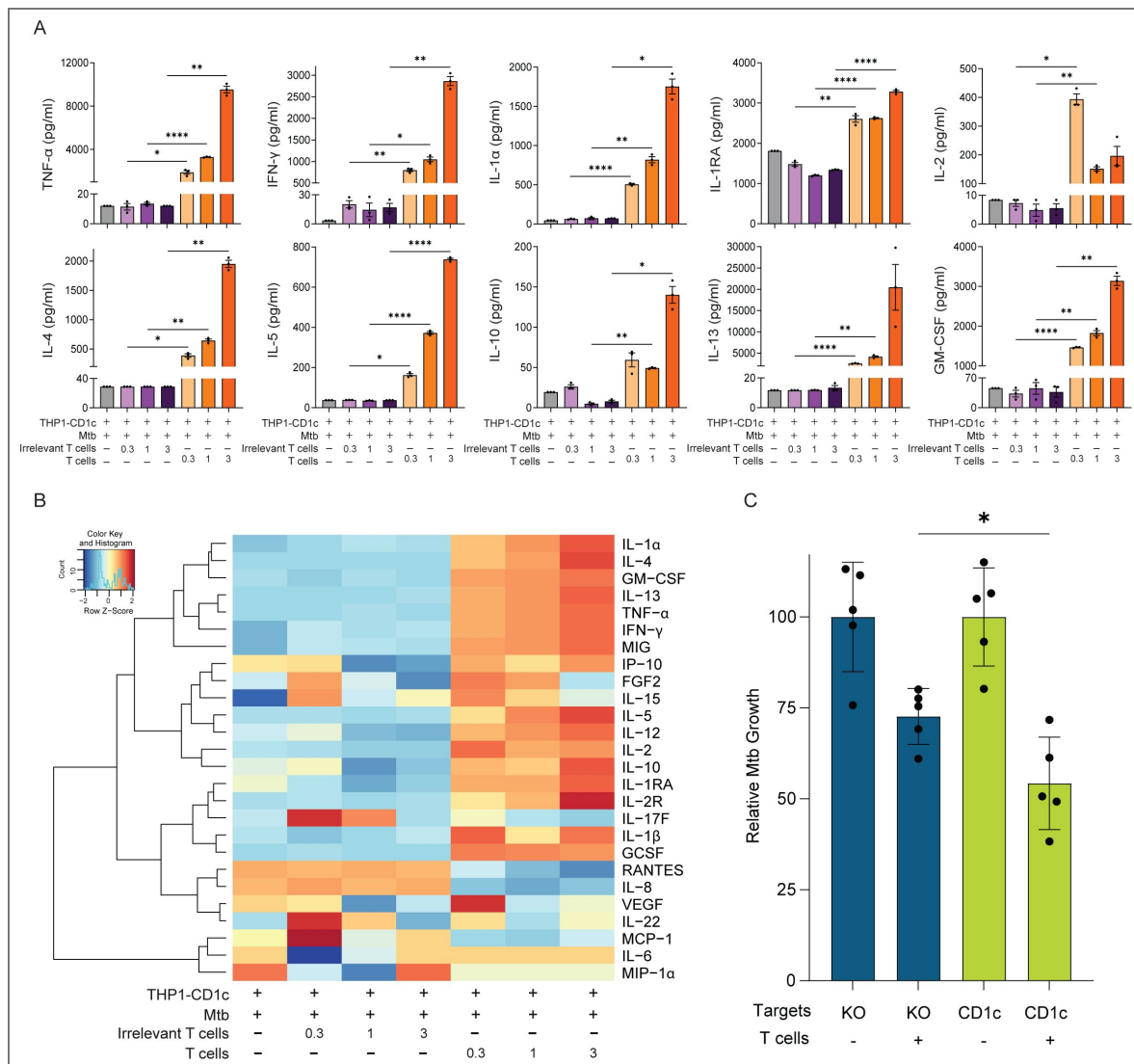


Figure 6. CD1c-autoreactive T-cells secrete diverse cytokines and reduce Mtb luminescence.

(A) Cytokines secreted by CD1c-autoreactive T-cells cultured with Mtb-infected THP1-CD1c APCs. CD1c-autoreactive T-cells produced significant amounts of the Th1 cytokines TNF- α , IFN- γ , IL-1 α and GM-CSF, and the Th2 cytokines IL-4, IL-5, IL-10 and IL-13 in a T-cell dose dependent manner. Irrelevant control T-cells did not release cytokines. (B) Heat map summarising cytokines released by CD1c-autoreactive T-cells, or irrelevant control T-cells (CD1c unrestricted), in response to Mtb-infected THP1-CD1c APCs. Red indicates high concentrations, and blue indicates low concentrations. Cytokine secretion was measured using a Luminex assay. Data are representative of two independent experiments, each performed in triplicate. (C) CD1c-autoreactive T-cells reduce Mtb luminescence significantly when cultured with THP1-CD1c APCs relevant to when they are cultured with THP1-KO APCs. T-cells cause some reduction in THP1-KO cells, and this is greater in THP1-CD1c cells. * $P < 0.05$; ** $P < 0.01$, **** $P < 0.0001$ (A, Two-way ANOVA, C, unpaired t-test).

Importantly, Mtb growth was significantly further reduced in infected THP1-CD1c APCs, consistent with a CD1c-dependent component of growth restriction, as luminescence closely correlates with colony-forming units⁴² (Fig. 6C). Overall, our findings demonstrate that CD1c-autoreactive T-cell responses are significantly greater towards Mtb-infected APCs than the basal autoreactive response and reduce relative Mtb burden under these *in vitro* conditions.

CD1c-autoreactive T-cells exhibit cytotoxic effector memory phenotype

Given their ability to restrict Mtb growth and secrete proinflammatory cytokines in response to infected APCs, we performed unbiased transcriptomic profiling to gain deeper insight into the functional properties of CD1c-autoreactive T-cells. To this end, we isolated T-cells *ex vivo* from two donors using barcoded “dCODE” CD1c-endo dextramers and performed single-cell RNA sequencing. CD3⁺ T-cells were enriched from PBMCs, and dextramer-positive and -negative T-cells were sorted for comparative analysis (Fig. 7A and Fig. S7). UMAP visualisation of the combined dataset revealed diverse T-cell populations, including naïve and effector memory (TEM) CD4⁺ and CD8⁺ T-cells (Fig. 7B and Fig. S8).

We focused on high-confidence CD1c-autoreactive T-cells, defined by co-positivity for CD1c-endo dextramer and dCODE barcode, and identified 2,117 such cells across both donors (Fig. S9). These cells were predominantly enriched in cytotoxic CD4⁺ and CD8⁺ effector memory clusters (Fig. 7C and 7D), although they also displayed transcriptional diversity spanning multiple T-cell states present in the broader dataset. Phenotypic analysis showed high expression of cytotoxic mediators including *GZMA*, *GZMK*, and *GNLY*, as well as effector molecules such as *CCL5* and *NKG7* (Fig. 7E). Pathway enrichment analysis revealed significant upregulation of gene sets related to response to other organisms, T-cell-mediated cytotoxicity, and cell killing (Fig. 7F). Together, these findings indicate that CD1c-autoreactive T-cells exhibit cytotoxic effector memory phenotypes and express key antimicrobial effector molecules, providing mechanistic insight into their potential effector function against Mtb-infected APCs.

Discussion

The majority of human CD1c-restricted T-cells are activated by exposure to CD1-expressing APCs in the absence of foreign lipid antigens^{23,24}. Indeed, such CD1c-autoreactive T-cells comprise 2% of all circulating human αβ T-cells²³, and studies have suggested a functional role for these T-cells in human autoimmune disease²⁹ and cancer²⁷. However, for almost the entirety of human evolution, infection is likely to have exerted a greater selective pressure affecting survival and reproduction than autoimmune disease or cancer^{43,44}. Here, we found that polyclonal CD1c-autoreactive T-cells produced higher levels of Th1 and Th2 cytokines when stimulated with Mtb-infected APCs relative to uninfected cells, mediated increased cytotoxicity against Mtb-infected APCs in a CD1c-mediated manner, and reduced the intracellular Mtb burden under *in vitro* conditions⁴². To gain insight into the cellular programmes associated with this antimicrobial effector potential, we performed single-cell RNA sequencing of *ex vivo* isolated CD1c-autoreactive T-cells from healthy donors. These cells were transcriptionally diverse but predominantly exhibited cytotoxic effector memory phenotypes. Notably, they expressed high levels of cytolytic molecules including *GZMA*, *GZMK*, and *GNLY*, as well as effector mediators such as *CCL5* and *NKG7*, consistent with a role in direct target cell killing⁴⁵. Granulysin is one of the few T-cell-derived molecules with demonstrated antimicrobial activity against Mtb⁴⁵, yet few human T-cell subsets express it in this context. Combined with our finding that CD1c-autoreactive T-cells can reduce intracellular Mtb luminescence *in vitro*, this places them among a rare population with antimicrobial activity in this experimental setting. These findings provide a mechanistic explanation for how CD1c-autoreactive T-cells could mediate cytotoxicity and suppress intracellular Mtb, supporting their classification as functionally distinct effector T-cells rather than merely autoreactive bystanders. To our knowledge, this is the first single-cell transcriptional analysis of human CD1c-autoreactive T-cells, offering preliminary insight into their phenotype and potential functional diversity. Our data suggest that CD1c-responsive T-cells, previously described

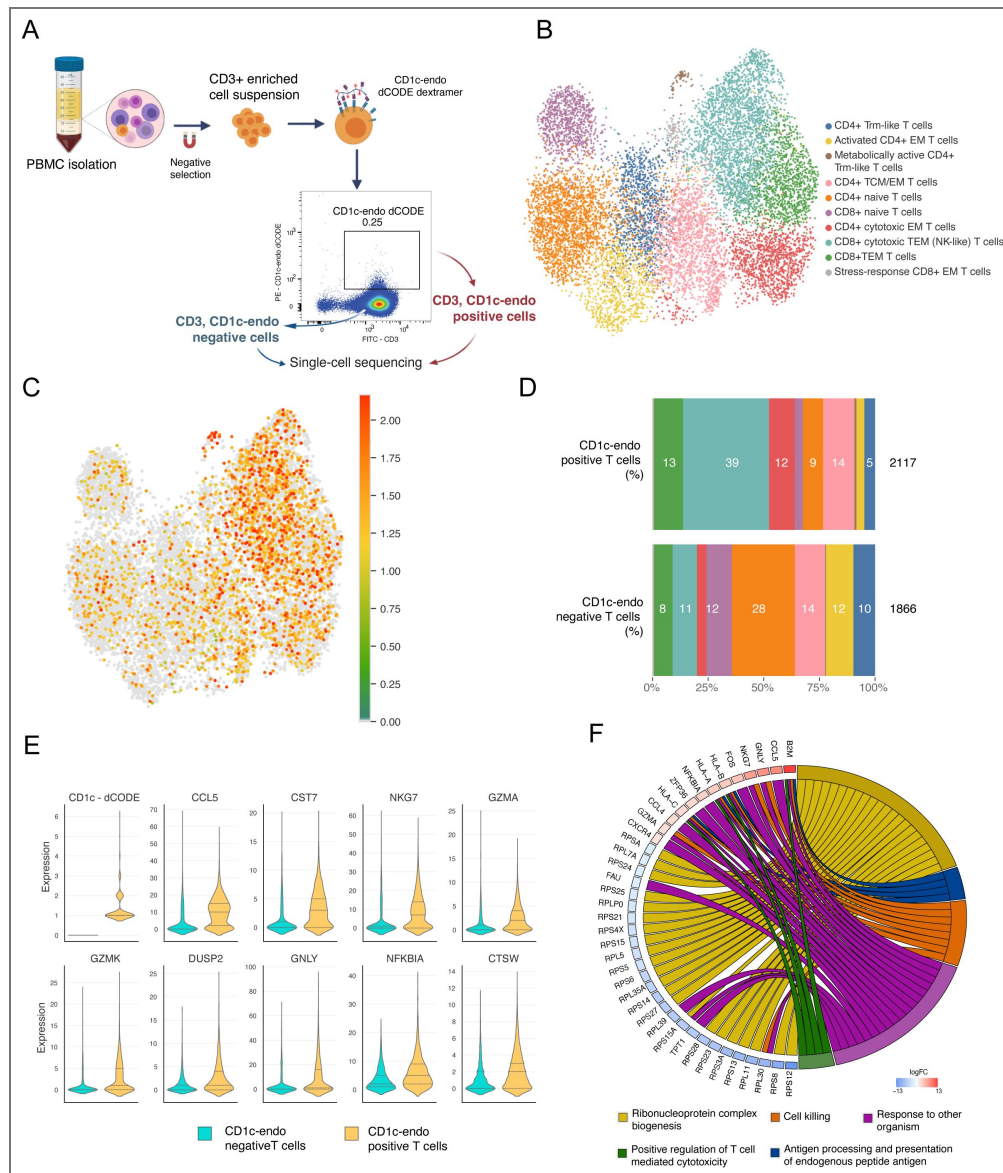


Figure 7. Single-cell profiling reveals that CD1c-endo dextramer positive T-cells are enriched for cytotoxic and effector phenotypes.

(A) Schematic overview of the sample processing workflow. PBMCs were isolated from two donors and CD3⁺ T-cells were subsequently enriched by negative selection. Cells were stained with CD1c-endo dCODE dextramer, and CD3⁺CD1c-endo⁺ (positive) and CD3⁺CD1c-endo⁻ (negative) T-cells were sorted for single-cell RNA sequencing. (B) UMAP visualisation of 11,804 single T-cells clustered by transcriptional profile. Clusters were annotated as functional T-cell subsets, including CD4⁺ and CD8⁺ naïve, central memory (TCM), effector memory (EM and TEM), cytotoxic, and stress-response populations. Additional subsets included metabolically active T cells and tissue-resident memory-like (Trm-like) CD4⁺ cells. TRM: tissue-resident memory; EM: effector memory; TEM: T effector memory; TCM: T central memory. (C) UMAP projection showing CD1c-endo dextramer binding intensity, reflecting the number of bound dextramer molecules per cell. Cells with higher binding intensities are enriched within cytotoxic and effector CD4⁺ and CD8⁺ subsets. (D) Bar plots showing the proportional distribution of T-cell subtypes among CD1c-endo-positive (top) and - negative (bottom) populations. CD1c-endo-positive T-cells were enriched for cytotoxic and effector subsets. (E) Violin plots showing expression levels of the top 10 differentially expressed genes by Wilcoxon rank-sum test. CD1c-endo positive T-cells upregulate genes associated with cytotoxicity (e.g., NKG7, GZMA, GZMK, GNLY, CTSW) and inflammation (e.g., CCL5, NFKBIA, DUSP2), consistent with a distinct effector phenotype. (F) Chord plot illustrating functional enrichment of differentially expressed genes in CD1c-endo positive T-cells. Upregulated genes are associated with biological processes such as cell killing, antigen processing and presentation, and response to other organisms. In contrast, downregulated genes predominantly map to ribonucleoprotein complex biogenesis. The colour gradient indicates the log fold-change (logFC) in gene expression.

as “autoreactive,” respond more robustly to Mtb-infected APCs, raising the possibility that, in addition to roles in autoimmunity and cancer, these cells may also contribute to antimicrobial responses.

Two previous studies utilising either *in vitro* activation-based assays or CD1c-endo tetramers have confirmed the presence of CD1c-autoreactive T-cells in human peripheral blood ^{23,46}. Although there was a discrepancy in their estimation of *ex vivo* frequencies for CD1c-autoreactive T-cells, both studies showed that these T-cells are a natural component of the human T-cell pool ^{23,46}. Moreover, both showed that CD1c-autoreactive T-cells were enriched for CD4⁺ or DN T-cells, and exhibited memory and adaptive T-cell features closely aligned with conventional MHC-restricted T-cells ^{23,46}. CD1c-autoreactive T-cells expressed diverse TCRs, including $\alpha\beta$, $\gamma\delta$ and $\delta/\alpha\beta$ ⁴⁶. Although the TCR repertoire was generally diverse, they were enriched for the TRAV17, TRAV38-1 and TRBV4-1 gene segments ^{46,47}. Functionally, CD1c-autoreactive T-cell clones exhibit a Th1- or a Th0-like phenotype, releasing IFN- γ , TNF- α , GM-CSF, IL-4, and IL-5 ²³. Corroborating these earlier studies, we detected CD1c-autoreactive T-cells in the blood of healthy donors, both $\alpha\beta$ and $\gamma\delta$, and demonstrated that CD1c-autoreactive T-cells were polyfunctional.

We generated CD1c-restricted $\alpha\beta$ T-cell lines with a baseline autoreactive response that was significantly greater when responding to Mtb-infected cells, with increased activation markers, cytotoxicity and cytokine secretion, and an associated reduction in Mtb luminescence in infected APCs. The Jurkat T-cell experiments with EM1 and EM2 TCRs support a key role for the TCR in mediating response through direct recognition of CD1c. These findings suggest that changes in CD1c expression or cytokine secretion alone are unlikely to explain the enhanced T-cell activity observed towards Mtb-infected cells and instead point to a role for lipid antigen presentation on CD1c. Our findings align with a study in 2005 where autoreactive CD1 group 1 restricted T-cell clones were generated by limited dilution, after stimulation of CD4-depleted T-cells with MoDCs pulsed with hydrophobic microbial extracts derived from Mtb, *E. coli*, and *Y. enterocolitica* ⁴⁸. A panel of 15 autoreactive clones were generated with the majority being CD1a- and CD1b-restricted and only two clones being CD1c-autoreactive ⁴⁸. While the clones exhibited weak autoreactivity, the response to microbial antigens far exceeded the self-reactive response seen in assays of T-cell proliferation ⁴⁸, similar to our findings and suggesting an antimicrobial role. TCR transfer experiments supported a role for the TCRs in promoting this dual reactivity to self and foreign antigen. However, further data from clone Mt2.33, which was a CD1c-autoreactive clone generated through Mtb antigen stimulation, was not provided. Further evidence for CD1c-restricted dual reactivity comes from a study identifying V δ 1⁺ $\gamma\delta$ T cells using CD1c-PM tetramers, where Jurkat cells transduced with these $\gamma\delta$ TCRs exhibited spontaneous activation and stronger responses to microbial lipid antigens, again implicating TCR-driven recognition of CD1c-presented ligands ⁴⁹. Another study investigated the CD1b-autoreactive clone HJ1, which was isolated from hCD1Tg mice, in *Listeria monocytogenes* infection ³³. While HJ1 T-cells exhibited autoreactivity towards CD1b, responses were enhanced by treatment of APCs with *Listeria* or the TLR agonists Pam3Cys (TLR2) and LPS (TLR4) ³³. Together, these observations suggest that some autoreactive CD1-responsive T-cells may have evolved as anti-microbial effectors, but when studied experimentally without such stimuli, a residual autoreactive phenotype is observed.

TLR activation during infection of APCs promotes the accumulation of stimulatory self-lipid antigens, contributing to the enhanced activation of iNKT cells ⁴⁰. Indeed, treatment of APCs with LPS or bacteria induced the accumulation of stimulatory iNKT self-lipid agonists b-d-glucopyranosylceramides (b-GlcCer) ⁴⁰. While we found that Mtb infection of THP1-CD1c APCs promoted enhanced cytotoxicity by CD1c-autoreactive T-cells, this effect was not observed with TLR2 or TLR4 activation alone. However, we could not rule out the possibility that Mtb-mediated induction of stimulatory lipids occurs through a TLR2- or TLR4-independent mechanism. Mechanistically, the enhanced activation of autoreactive T-cells in our studies could be driven by recognition of Mtb lipids as suggested previously ^{48,49}, or by recognition of mammalian and bacterial shared lipids as shown for CD1b-autoreactive T-cells ^{34,50}. It is possible that Mtb infection of APCs facilitates the accumulation of stimulatory CD1c presented self-lipids, inducing the observed T-cell responses against Mtb-infected APCs. Stimulatory “stress lipids” may include

cholesteryl esters²⁶ or sequestered “permissive” lipid ligands that can be buried deeply within the CD1c groove, exposing the CD1c protein surface for direct recognition by the TCR²⁵, a hypothesis supported by our TCR transfer experiments. However, a limitation of this study is that we did not identify specific lipids presented by CD1c on Mtb-infected APCs, making it unclear whether they are of host or bacterial origin. Future studies should aim to characterise these lipids to better understand the mechanism underlying CD1c-mediated T-cell activation. Lipid identification would require isolation of CD1c molecules from Mtb-infected cells, followed by mass spectrometry analysis to determine the specific lipids involved⁵¹. In addition, our experiments rely on *in vitro* human cell systems, and *in vivo* validation of CD1c-restricted responses will benefit from specialised CD1-transgenic models that accommodate human CD1c biology, beyond what is possible in standard mouse systems^{32,52}.

Our observation of CD1c expression within human lung TB granulomas provides further evidence for a potential role for lipid-specific T-cell immunity in the host immune response to Mtb. Our findings are consistent with reports suggesting that CD1c expression is targeted by Mtb and BCG, leading to its reduced expression on APCs such as DCs^{36,37,53,54}. A recent study identified a potential mechanism, with Mtb driving expression of the microRNA miR-381-3p and thereby suppressing CD1c³⁷. Therefore, our data are consistent with the idea that CD1c-autoreactive responses contribute to protective immunity, and that Mtb suppresses CD1c expression at sites of infection as part of an immune evasion strategy.

Taken together, our findings suggest that T-cells previously described as autoreactive may also play a role in anti-mycobacterial immunity, and that this phenotype could have been shaped by evolutionary adaptation to mycobacterial exposure. Mtb has killed many more humans than any other single infection, and humans and Mtb may have co-evolved for 70,000 years^{55,56}, demonstrating the significant impact mycobacteria will have had on human immune development. We identify functional responses to Mtb-infected cells by CD1c-autoreactive T-cells, including activation, cytotoxicity, cytokine secretion and reduction in Mtb luminescence *in vitro*, that clearly exceed the autoreactive phenotype. Together, our findings suggest that CD1c-autoreactive T-cells may contribute to immune responses against infection and warrant further investigation in the context of vaccine development.

Methods

Flow cytometry

The following fluorescent reagents were used: All antibodies were from Biolegend unless stated; anti-CD3-Bv510 (clone UCHT1); anti-CD3-FITC (clone UCHT1); anti-MHCI-PE (clone W6/32); anti-MHC-II-FITC (clone Tu39); anti-β2m-PerCP-Cy5.5 (clone A17082A); anti-αβTCR-Bv421 (clone IP26); anti-γδTCR-APC (clone B1); anti-CD4-FITC (clone RPA-T4); anti-CD8-APC (clone HIT8a); anti-CD14-FITC (clone M5E2); anti-CD11c-Bv421 (clone Bu15); anti-HLA-DR-PE/Cy7 (clone L243); anti-CD1a-APC (clone HI149); anti-CD1b-APC (clone SN13(K5-1B8)); anti-CD1c-APC (L161); anti-CD1c-PE (clone L161); anti-CD69-PE (clone FN50); anti-CD25-PE/Cy7 (clone BC96); anti-CD137-APC (clone 4B4-1); LIVE/DEAD Fixable Near-IR Dead Cell Marker and Cell Trace Violet (CTV) (both ThermoFisher Scientific); and Propidium iodide (Sigma). T-cells were treated with 50 nM Dasatinib (Axon) and blocked with Human TruStain FcX™ (Biolegend) for 30 minutes at 37°C, and anti-CD36 (clone 5-271, Biolegend) for 20 minutes before the addition of tetramers and staining reagents. Cells were stained for 45 minutes, extensively washed and acquired on a FACSARIA Fusion, or a FACSARIA II (All BD Biosciences). Data was analysed using FlowJo VX software (FlowJo LLC).

Cell lines

The following cell lines were utilised in this study: THP1 (myelomonocytic leukemia), J.RT3-T3.5 (TCRβ-deficient T-cell leukemia) and HEK293TN (Human Embryonic Kidney). The engineered THP1-KO cells were generated using CRISPR-Cas9 technology to prevent the expression of β2m and Class II transactivator (CIITA). Following the knockout of β2m and CIITA, the cells were transduced using a lentivirus system with a plasmid containing a β2m-CD1c single chain gene construct to

generate the THP1-CD1c cell line. Three days post transduction, CD1c⁺ cells were sorted by flow cytometry and maintained in culture. Jurkat, THP1 and HEK293TN cells were maintained in complete RPMI or DMEM (for HEK293TN) containing 10% Foetal Bovine Serum (Sigma), 1% Non-essential Amino Acids, 1% L-glutamax, 1% Sodium Pyruvate, 1% Penicillin/Streptomycin (all from ThermoFisher Scientific). All primary cell lines used in this study were isolated from peripheral blood mononuclear cells (PBMCs) obtained using Ficoll-Paque (Cytiva) from healthy blood bank donors and maintained in culture in T-cell media; RPMI containing 5% Human AB Serum (Merck), 1% Penicillin/Streptomycin, 1% L-glutamax, 1% Non-essential Amino Acids, 1% Sodium Pyruvate Pyruvate (All ThermoFisher Scientific). Pan T-cells from healthy donors purified by EasySep Human T-cell Isolation Kit (Stem cell Technologies) were labelled with CTV according to manufacturer instructions and stimulated with irradiated (80 Grays) THP1-KO or THP1-CD1c cells at a 6:1 ratio, or without THP1 cells as control. On days 4, 6, 8, 10 and 12 post stimulation, 10 IU/ml rhIL-2 (Proleukin, Chiron) was added. At 12 days post stimulation, cells were rechallenged with THP1-KO or THP1-CD1c APCs overnight before staining for activation marker expression on live CD3⁺ CTV low cells by flow cytometry. Some cultures were stained with CD1c-endo tetramers directly on day 12 and then tetramer positive T-cells were sorted by flow cytometry and expanded in the presence of phytohemagglutinin (PHA) (ThermoFisher Scientific) (1µg/ml), rhIL-2 (100U/ml), and irradiated PBMC (5×10⁵ cells/ml).

Tetramers

Human CD1c was produced as a single chain construct with β2m N-terminally fused to the heavy chain via a flexible glycine-serine linker. Expression cassettes were cloned in-frame with C-terminal Avi-tag and His6 tag in pCDNA3.1 encoding an additional BirA Ligase. Protein expression was performed using the Expi293 Expression System (ThermoFisher Scientific). Soluble CD1c proteins were purified using nickel affinity and size exclusion chromatography. For CD1c-endo tetramers, purified CD1c-endo monomers were used to generate fluorescent-labelled CD1 tetramers by conjugation to PE-streptavidin (Biolegend). To generate dextramers, the same process was repeated as above, with PE-labelled dextran backbones (Dex-PE) (Immudex).

TCR sequencing

Single cells were isolated by CD1c-endo tetramer guided cell sorting into a 96 well plate. Cells were lysed and an Oligo-dT primer was annealed to the Poly A tails of the mRNA, and reverse transcription was carried out to generate cDNA. During reverse transcription, unique adaptor PCR arms were added to the cDNA, primers that anneal to the unique PCR adaptor arms were then added to allow for the universal PCR amplification of all cDNA. Targeted PCRs were then preformed for each single cell to amplify the variable regions of TCRs, the products were barcoded with an 8-bp DNA sequence unique to each well. The PCR products were then purified and DNA fragments of 500-600bp were targeted. NGS libraries of the size-selected fragments were then prepared and barcoded using kits (New England Biolabs) and sequenced using the MiSeq platform. Sequences of the CDR3, variable, and constant regions were obtained. The TCR-sequencing was then further analysed using Seven Bridges software to generate the final sequences.

TCR cloning and transduction

Sequences identified from tetramer-guided sorting were constructed into full length TCR, synthesised and sub-cloned into the pELNS Lentivector by Genscript. Adherent HEK 293TN cells were co-transfected with pELNS lentivector (2.5 µg) and three accessory plasmids: pCMV-VSV-G (1.5 µg), pRSV.REV (3 µg), and pMDL.pg.RRE (3 µg), to induce production of Lentiviral particles. Lentiviral particles were harvested, filtered, and used directly for transduction of β2m knock-out J.RT3-T3.5 Jurkat cells (kind gift from Pierre Vantourout, King's college London). Transduced cells were sorted by flow cytometry on a FACSARIA IIIU (BD Biosciences).

CD1c-endo streptamers

Strep-Tactin® magnetic microbeads (IBA Lifesciences) were prepared by washing and resuspending in binding buffer (PBS, 1mM EDTA, 0.5% bovine serum albumin) prior to mixing with CD1c-endo incorporating Strep•Tag® II peptide sequence and incubated at 4°C overnight. The following day, additional CD1c-endo was added and incubated for 30 minutes at 4°C to complete the preparation of the CD1c-endo streptamer. To enrich CD1c-restricted T-cells, 2×10^7 T-cells were resuspended in binding buffer containing 50nM dasatinib (Sigma) containing Human TruStain FcX™ (Biolegend) and incubated at room temperature for 10 minutes. CD1c-endo streptamer was then added to T-cells and incubated on ice for 20 minutes before separating the CD1c-endo streptamer bound cells (positive fraction) and unbound cells (negative fraction) by magnetic sorting. The enrichment procedure was repeated with the negative fraction. The positive fractions were next pooled and thoroughly washed. To remove bound T-cells in the positive fraction from the CD1c-endo streptamer, the T-cells were resuspended in biotin elution buffer and incubated on ice for 10 minutes and separated using a magnet three times. The positive and negative fractions were stained and analysed by flow cytometry. CD1c-endo dextramer stained cells from the positive fraction were sorted by flow cytometry into a single well of a 96-well plate containing T-cell media and expanded with 1µg/mL PHA, 100IU/mL rhIL-2 and 5×10^5 irradiated PBMCs from three different donors.

Mtb Culture

Mycobacterium tuberculosis H37Rv (Mtb H37Rv) was cultured in Middlebrook 7H9 medium (supplemented with 10% ADC, 0.2% glycerol and 0.02% Tween 80) (BD Biosciences) at 37°C in an incubator with shaking at 200 rpm. Bioluminescent Mtb H37Rv lux was cultured with kanamycin (25 µg/mL). Cultures at $\sim 1 \times 10^8$ CFU/ml Mtb (OD = 0.6) were used for all experiments.

MoDC generation

To generate MoDC, CD14⁺ monocytes were isolated from PBMCs by negative selection using magnetic beads (Stem cell Technologies) and cultured in the presence of 25ng/ml rhGM-CSF and 20ng/ml rhIL-4 (both from Miltenyi) for 5 days.

CD1 expression

MoDC

After 3 days of culturing monocytes in the presence of rhGM-CSF and rhIL-4, some wells were infected with live Mtb (MOI=1) whereas others were left uninfected. On Day 0, 3, 5 and 7, cells were collected from wells by thoroughly washing with 3mM EDTA and stained for flow cytometry analysis. Cells were acquired on FACS Aria IIU (BD Biosciences) to measure CD1a, CD1b and CD1c expression.

THP1 cells

THP1-KO and THP1-CD1c cells were stained with anti-CD1c-PE and its isotype control (BioLegend) and Live/Dead Aqua (Invitrogen). On day 0, some wells were infected with Mtb (MOI=1) whereas others were left uninfected. On Day 3, cells were collected from wells by thoroughly washing with 3mM EDTA and stained for flow cytometry analysis. Cells were acquired on FACS Aria IIU (BD Biosciences) to confirm the expression of CD1c on THP1 cells.

Immunohistochemistry

Tissue sections (4µm thick) were dewaxed and rehydrated before inhibiting endogenous peroxidase. Heat-induced epitope retrieval was performed prior to blocking non-specific staining. Primary antibodies, 1:500 anti-CD1c recombinant rabbit monoclonal (EPR23189-196, Abcam) was then applied to the sections and incubated overnight at 4°C. Sections incubated with buffer alone were included as a negative control. Secondary antibody, goat anti-rabbit 1:800 (2B Scientific) was applied. Sections were developed with avidin-biotin peroxidase complexes (2B Scientific) and 3,3'-

diaminobenzidine tetrahydrochloride (DAB; Launch Diagnostics). Sections were counterstained with Mayer's haematoxylin, dehydrated, cleared and mounted using XTF mounting medium (CellPath). Sections were allowed to dry before subsequent imaging using an Olympus Bx51, CC12 DotSlide microscope.

Differential Gene Expression Analysis

PRJNA478394 fastq files were downloaded from ENA using enaBrowserTools (v1.1.0), after which transcripts in each file were quantified against a human genome template (downloaded from ftp://ftp.ensembl.org/pub/release106/fasta/homo_sapiens/cdna/Homo_sapiens.GRCh38.cdna.all.fa.gz) using alignment-free software kallisto (v0.46.1). Counts were imported into R using tximport (v1.28.0) and converted to counts per million (CPM) using edgeR (v3.14.0). Genes with summed CPM values below 14 across five samples (corresponding to the number of donors) were removed from the expression matrix, which was then normalised using TMM normalisation. Differential gene expression analysis was performed using the voom-limma (limma v3.56.2) pipeline, where normalised counts were compared between (live) Mtb-infected MoDCs and uninfected MoDCs (controls) at 48 hours. Computed p-values were adjusted for multiple hypothesis testing using the Benjamini-Hochberg method, and differentially expressed genes ($> +1 \log_2\text{FC}$ = Upregulated, $< -1 \log_2\text{FC}$ = Downregulated with an adjusted p.value < 0.05) were subsequently identified.

Gene expression heatmap

Counts in the filtered, normalised expression matrix were converted to CPM, then $\log_2$ transformed. The expression matrix was subset to retain values for *CD1A*, *CD1B* and *CD1C* for the control and (live) Mtb-infected samples at 48 hours post-infection. Coolmap function from limma was used to generate a heatmap showing the relative change in gene expression for each sample, from the average expression for each gene, using “de pattern”. Columns were clustered using “average” method.

Activation assays

T-cells

T-cells were cultured with THP1-KO and THP1-CD1c cells that were either uninfected or infected with live Mtb (MOI=1) for 48 hours at 37°C. Activation status of T-cells was measured through CD69 and CD25 expression by flow cytometry on a FACSAria IIU (BD Biosciences).

Jurkat T-cells

TCR transduced J.RT3-T3.5 Jurkat T-cells were cultured for 20 hours at 37°C in 96-well plates coated with CD1c-endo protein monomers (10µg/mL). TCR transduced J.RT3-T3.5 Jurkat T-cells cultured alone served as negative control. TCR transduced J.RT3-T3.5 Jurkat T-cells were also cultured with THP1-KO and THP1-CD1c cells that were either uninfected or infected with live Mtb (MOI=1) for 20 hours at 37°C. Activation status of Jurkat T-cells was measured through CD69 expression by flow cytometry on a FACSAria IIU.

Activation Induced Marker (AIM) assay

Pan T-cells were isolated from healthy human blood using an EasySep Human T-cell Isolation Kit (Stem cell Technologies) and stained with CTV to track proliferation. Purified T-cells are stimulated with THP1-CD1c cells for 12 days as above. On day 12, T-cells were re-stimulated overnight with irradiated THP1-KO, THP1-CD1c or left unstimulated. Proliferated T-cells were determined as CTV low and activation was measured as upregulation of $\text{CD}25^+/\text{CD}137^+$ or $\text{CD}69^+/\text{CD}137^+$ on T-cells by flow cytometry.

Cytotoxicity assays

ToxiLight cytotoxicity assay

T-cell cytotoxicity against UV-killed Mtb-treated target cells was measured using a ToxiLight assay (Lonza), which was carried out as per the manufacturer's instructions. Briefly, 20 µl of media from each well was transferred to a luminometer compatible 96-well plate, 100 µl of ToxiLight reagent was then added. Following a 5-minute incubation period, the plate was read using a Glomax 20/20 Luminometer (Glo-Max Discover; Promega).

Lysis assay

Tag-it Violet (Biolegend) stained 5×10^4 THP1-KO or THP1-CD1c cells were left either uninfected or infected with live Mtb (MOI=1) or stimulated with either 1 µg/mL LPS (Sigma) or 1 µg/mL Pam3CSK4 (InvivoGen) for 48 hours. These cells were then co-cultured with 2.5×10^3 CD1c-autoreactive T-cells or cultured alone for 48 hours, after which, cells were collected, stained with fixable live/dead stain and analysed by flow cytometry.

Luminex xMAP assays

T-cells were re-stimulated overnight with irradiated THP1-KO, or THP1-CD1c cells or left unstimulated. Additionally, CD1c-autoreactive T-cells, or control irrelevant T-cells, were co-cultured with untreated or UV-killed Mtb (MOI 10) treated THP1-KO and THP1-CD1c cells. Mtb untreated or treated THP1 cells and T-cells were cultured alone as control. At 24 hours after stimulation, supernatant was removed from each T-cell co-culture for multiplex cytokine analysis, cells were pelleted and the supernatant frozen. We followed manufacturer's protocol to determine concentrations of cytokines (Life Technologies). A Bioplex 200 platform (Bio-Rad) was used to measure concentrations of cytokines in a multiplex panel: GM-CSF, IFN γ , IL-1 β , IL-10, IL-12p70, IL-13, IL-17a, IL-2, IL-22, IL-4, IL-5, IL-8, IP-10, MIP-1 α , MIP-1 β , Oncostatin M, RANTES and TNF α or the cytokine 35-plex human panel (Thermo Fisher Scientific).

Mtb Lux growth assays

THP1-KO and THP1-CD1c were infected with Mtb lux (MOI=1). After overnight infection, cells were transferred from vented flasks to 50 mL Falcon tubes after detachment with Versene solution (Sigma Aldrich) for 10 min. Cells were then washed with Hanks' balanced salt solution (HBSS) without Ca/Mg (Thermo Fisher Scientific) and then centrifuged at $380 \times g$ for 5 minutes at 4°C to remove extracellular bacteria. THP1-KO and THP1-CD1c were next resuspended in T-cell media containing kanamycin (25 µg/mL) and co-cultured with T-cells and incubated at 37°C, 21% O $_2$ and 5% CO $_2$. Mtb growth was monitored by luminescence (GloMax 20/20 Luminometer, Promega) for 7 days. Relative luminescence values for each condition were normalised to the matched infected APC-only control within each experimental replicate, ensuring that baseline growth levels were accounted for independently in each experiment.

Single cell library preparation and sequencing

CD3-enriched PBMCs were washed in cold staining buffer (1% BSA in PBS), counted, and assessed for viability. One million cells per sample were resuspended in binding buffer containing 50 nM dasatinib (Sigma), blocked (as above) and then stained with PE-dCODE™ dextramers conjugated to CD1c-endo, in combination with the following fluorochrome-conjugated antibodies: FITC-conjugated anti-human CD3 (clone UCHT1) and viability dye (Live/Dead Near IR, Invitrogen, Basingstoke, UK). Following a 30-minute incubation at 4 °C, cells were washed twice with ice-cold PBS supplemented with 1% FBS and sorted on a BD FACSAria cell sorter (BD Biosciences). Sorted cell fractions were labelled with sample-specific TotalSeq™-C hashtag antibodies. Flow cytometry data were analysed using FlowJo v10.8.1 (FlowJo LLC). Sorted cells were loaded onto a 10x Genomics Chromium Chip G at a target of ~45,000 cells per lane using the Single Cell V(D)J Reagent Kit v1.1. First-strand cDNA synthesis was performed according to the manufacturer's protocol. After emulsion disruption and nucleic acid recovery, cDNA amplification proceeded following the Chromium Next GEM Single Cell V(D)J v1.1 user guide.

scRNA-seq alignment and analysis of CD1c-endo dextramer-sorted T-cells

Sequencing reads were de-multiplexed and aligned using Cell Ranger v7.2.0, with mapping to the GRCh38 reference genome (build 2020-A, 10x Genomics). A total of 14,524 single CD3⁺ T-cells were initially profiled by scRNA-seq following dextramer-guided cell sorting. Cells were sorted into CD3⁺CD1c-endo-positive and CD3⁺CD1c-endo-negative fractions using dCODE™ dextramer reagents. Raw gene expression matrices were generated and processed for downstream analysis.

Quality control and filtering

Cells underwent rigorous quality control (QC) to ensure data integrity and remove low-quality or ambiguous profiles. Specifically, the following metrics were calculated per cell: number of detected genes, total unique molecular identifiers, mitochondrial gene content, and ribosomal gene content. Additionally, two independent doublet detection methods were applied: Scrublet, and scDblFinder. Outlier cells were excluded based on the following criteria: low gene content and Unique Molecular Identifier (UMI) counts, high mitochondrial content (>10%) and high ribosomal content (>50%). Predicted doublets from at least one of the two detection tools. After filtering, a total of 11,804 high-quality single T-cells were retained for downstream analysis.

Annotation of CD1c-endo binding

Cells were annotated as CD1c-endo-positive or -negative based on dextramer binding. Of the 11,804 that passed quality control, 9,518 lacked detectable dextramer binding and were initially labelled as CD1c-endo negative. These were further stratified by sorting origin: 7,652 cells originated from the CD1c-endo-positive sort gate, and 1,866 cells came from the CD1c-endo-negative sort gate. Only the 1,866 cells from the CD1c-endo-negative sort gate were retained as confidently assigned CD1c-endo-negative T-cells. The remaining 7,652 cells, despite lacking dextramer signal, were excluded due to possible false negatives introduced during staining or sorting. To accurately classify CD1c-endo-positive cells, a density-based thresholding approach was applied. The distribution of dextramer signal was modelled using kernel density estimation (KDE). To define confidently positive cells, thresholds were set based on the 1st and 99th percentiles and the interquartile range (IQR) of dextramer intensity. Cells with signal below the lower bound ($Q1 - 1.5 \times IQR$) were excluded as potential noise or non-specific binders. This approach minimised false positives and enabled robust separation of CD1c-endo-positive and -negative populations. After applying quality control filters and dextramer-based gating, CD1c-endo-positive T-cells were defined as those confidently binding dextramer above the KDE threshold, while CD1c-endo-negative T-cells comprised the 1,866 cells from the negative gate lacking any detectable dextramer signal.

Subsampling analysis to assess robustness

To control for group size imbalance (CD1c-endo⁺ vs CD1c-endo⁻), we performed a subsampling analysis. From each group, a random subset of cells equal to the smallest group size (1,866) was sampled using `np.random.choice` (without replacement). This balanced dataset was reanalysed to ensure that key differential gene expression patterns and clustering results were consistent across equal-sized groups. The findings from the full dataset were confirmed, indicating that group size imbalance did not bias the results, validating the robustness of the conclusions.

Study approval

All clinical studies were conducted according to the Declaration of Helsinki principles. All participants gave written informed consent. Paraffin-embedded lung tissue from TB patients or those with adenocarcinoma were retrieved from the histology archive at University Hospital Southampton with approval by the Institutional Review Board (Reference 12/NW/0794: SRB04_14). The ethics committee approved immunohistochemical analysis without individual informed consent since it was surplus archived tissue taken as part of routine care.

Statistical Analysis

GraphPad Prism version 10 (GraphPad Software, Inc.) was used for statistical analysis, and P values ≤ 0.05 were considered statistically significant. The Mann–Whitney U test, T tests, one-way ANOVA, Wilcoxon signed-rank test, 2-way ANOVA and Tukeys test were used as stated in the figure legends.

Supplementary Figures

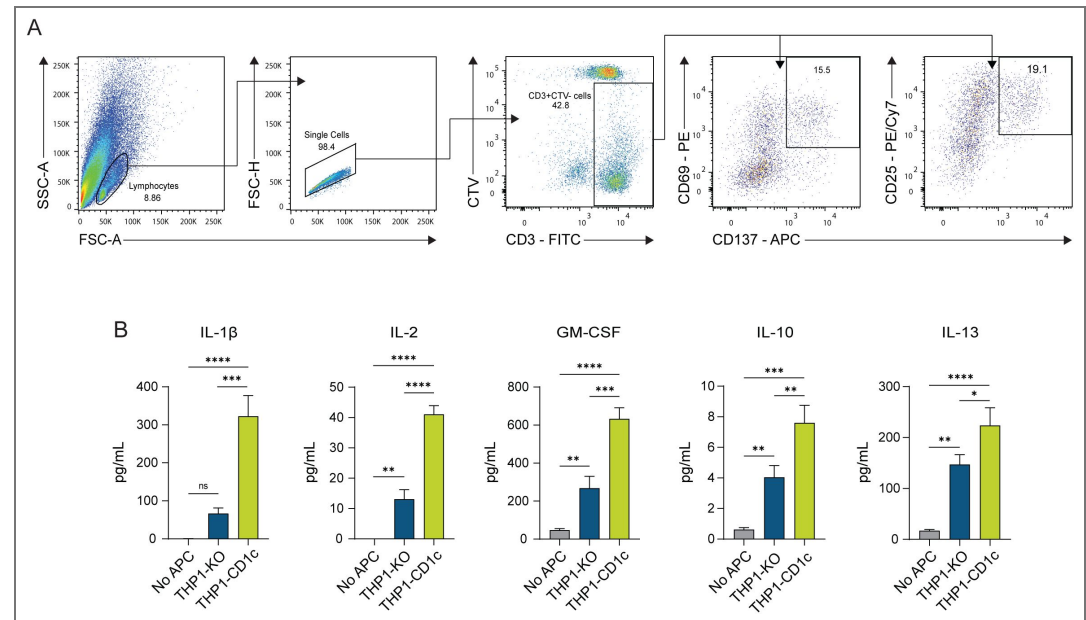


Figure S1. (A) Flow cytometry gating strategy analysing the expression of the T-cell activation markers CD69, CD25, and CD137 on proliferated CD3⁺CTV⁺ T-cells. (B) Cytokine release by T-cells first expanded with THP1-CD1c APCs and then stimulated overnight with THP1-KO or THP1-CD1c APCs. Cytokine secretion was measured by Luminex array. * P < 0.05; ** P < 0.01, *** P < 0.001, **** P < 0.0001 (one-way ANOVA with Tukey's multiple comparison test). Mean and SD of triplicate measurements are shown and are representative of three individual donors.

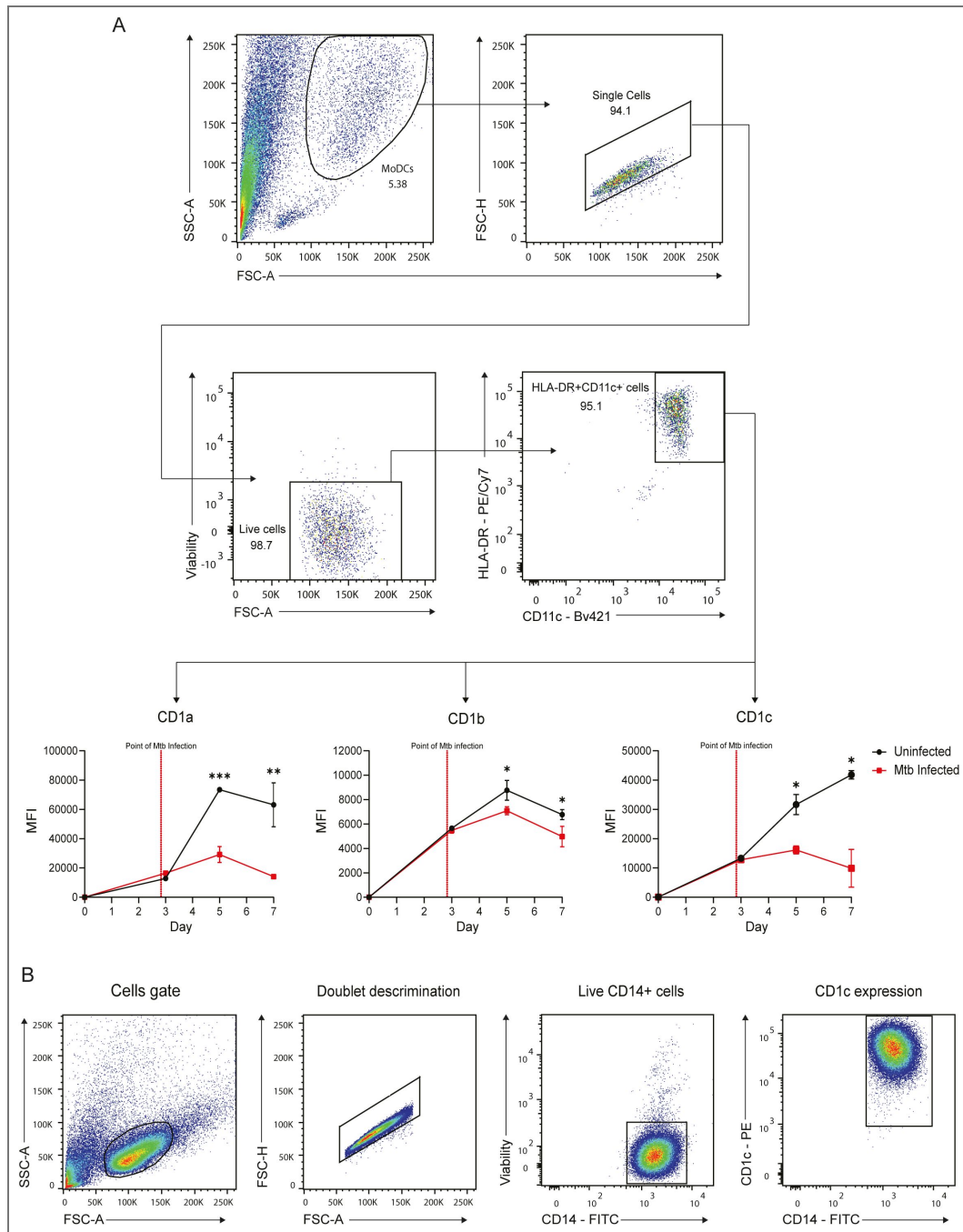


Figure S2.

(A) Flow cytometry gating strategy of live HLA-DR⁺/CD11c⁺ MoDCs (top), and line graphs showing the effect of Mtb infection on the expression of CD1a, CD1b, and CD1c on MoDCs (bottom). (B) Flow cytometry gating strategy depicting live CD14⁺ THP1-CD1c cells stained with anti-CD1c antibody.

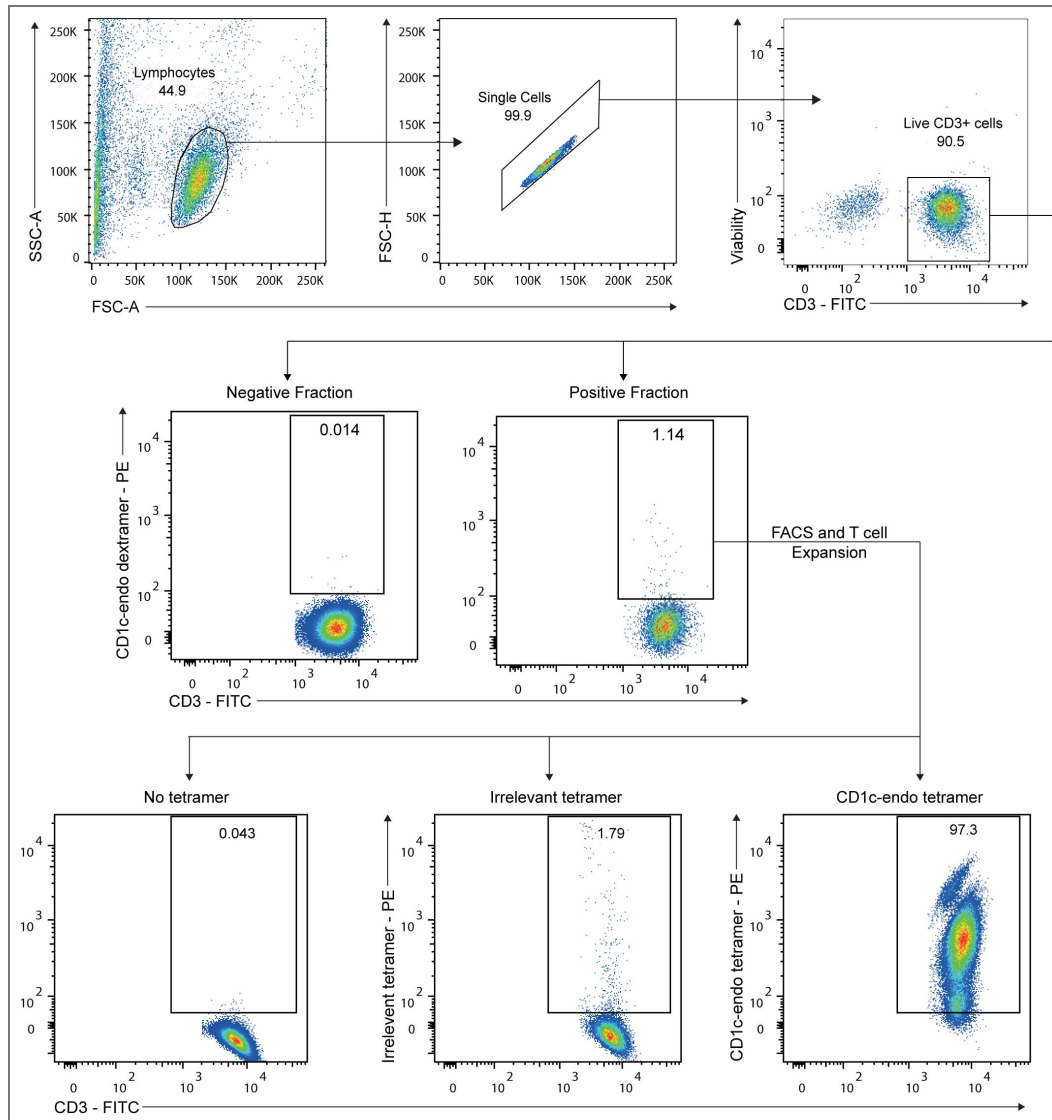


Figure S3. Flow cytometry gating strategy depicting live CD3⁺ T-cells comprising the negative (cells that did not bind the streptamers) or the CD1c-endo streptamer positive T-cell fraction (containing 1.14% CD1c-endo positive T-cells) stained with CD1c-endo dextrans.

CD1c-endo dextramer positive T-cells were sorted and expanded. After expansion, T-cells were either unstained, or stained with an irrelevant tetramer or with CD1c-endo tetramer. Enriched T-cells brightly stain with CD1c-endo tetramers.

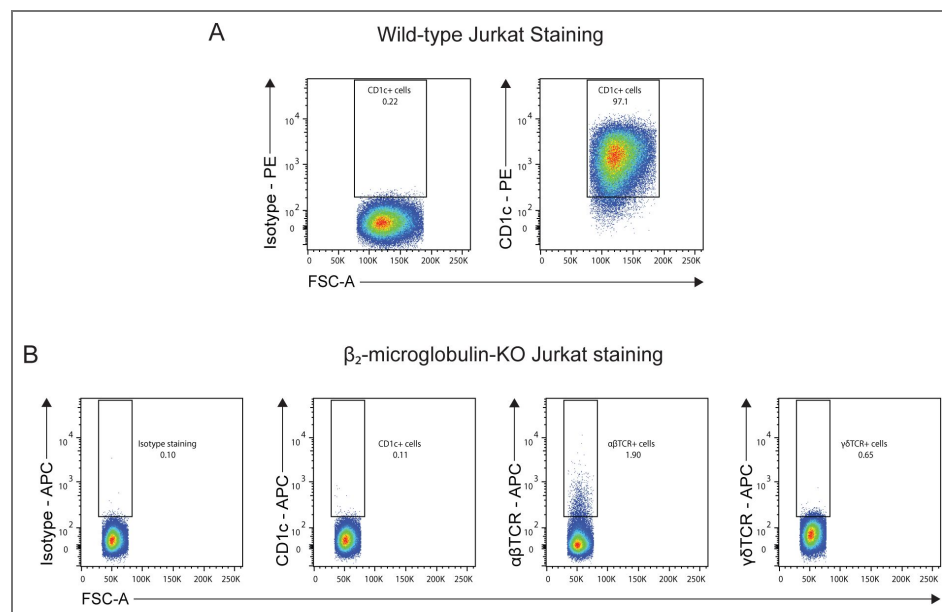


Figure S4.

(A) Flow cytometry dot plots showing high expression of CD1c on wild-type JRT3.5 Jurkat T-cells. (B) Flow cytometry dot plots showing the absence of CD1c and TCR expression on β₂-microglobulin knock-out JRT3.5 Jurkat T-cells.

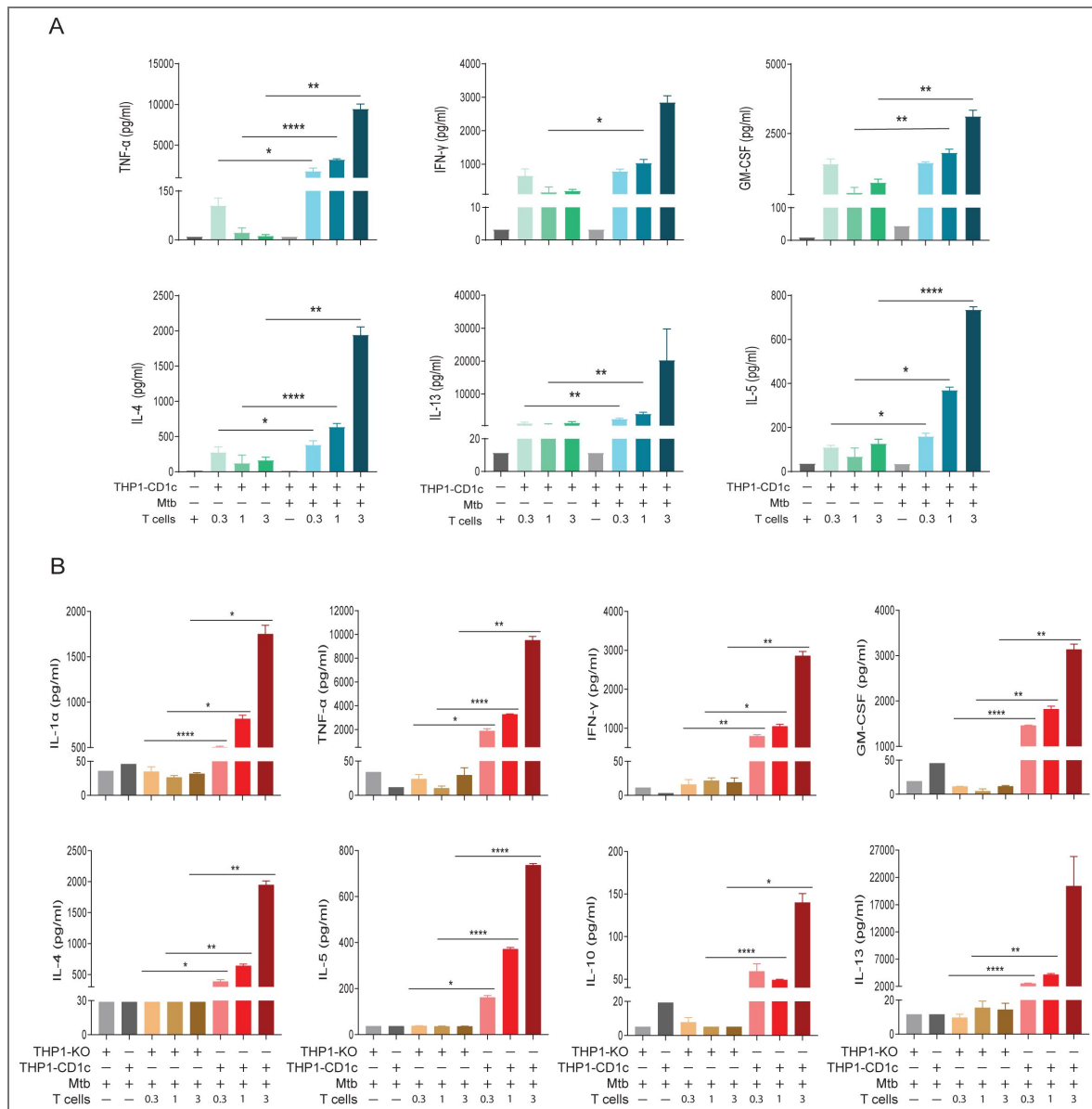


Figure S5.

(A) CD1c-autoreactive T-cells secrete cytokines in response to untreated THP1-CD1c APCs in an autoreactive manner. Importantly, CD1c-autoreactive T-cells release significantly higher concentrations of cytokine when cultured with Mtb-infected THP1-CD1c APCs. (B) CD1c-autoreactive T-cells secrete diverse cytokines in a CD1c dependent manner. CD1c-autoreactive T-cells release cytokines when cultured with Mtb-treated THP1-CD1c APCs but not when were cultured with Mtb-treated THP1-KO APCs. Data are representative of two independent experiments, each preformed in triplicate. * $P < 0.05$; ** $P < 0.01$, **** $P < 0.0001$ (Two-way ANOVA).

Figure S6. UV Mtb-treated THP1-CD1c APCs secrete chemokines IL-8 and RANTES (pink bars), in comparison to media only (black bars), and untreated THP1-CD1c APCs (blue bars).

Data are representative of two independent experiments, each preformed in triplicate. **** P < 0.0001 (Two-way ANOVA).

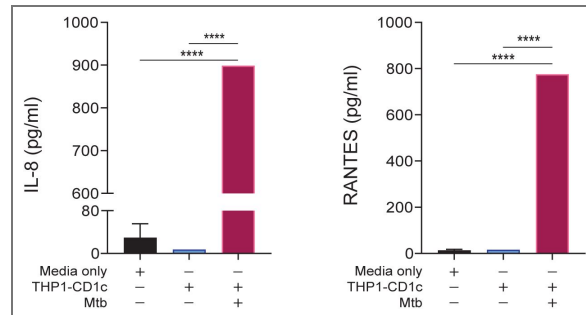


Figure S7.

(A) Representative flow cytometry gating strategy used to isolate CD3⁺ T-cells stained with CD1c-endo dCODE dextrans. (B) Representative plots from two donors showing distinct CD3⁺CD1c-endo⁺ and CD3⁺CD1c-endo⁻ T-cell populations. Percentages indicate the proportion of positive cells within the CD3⁺ gate.

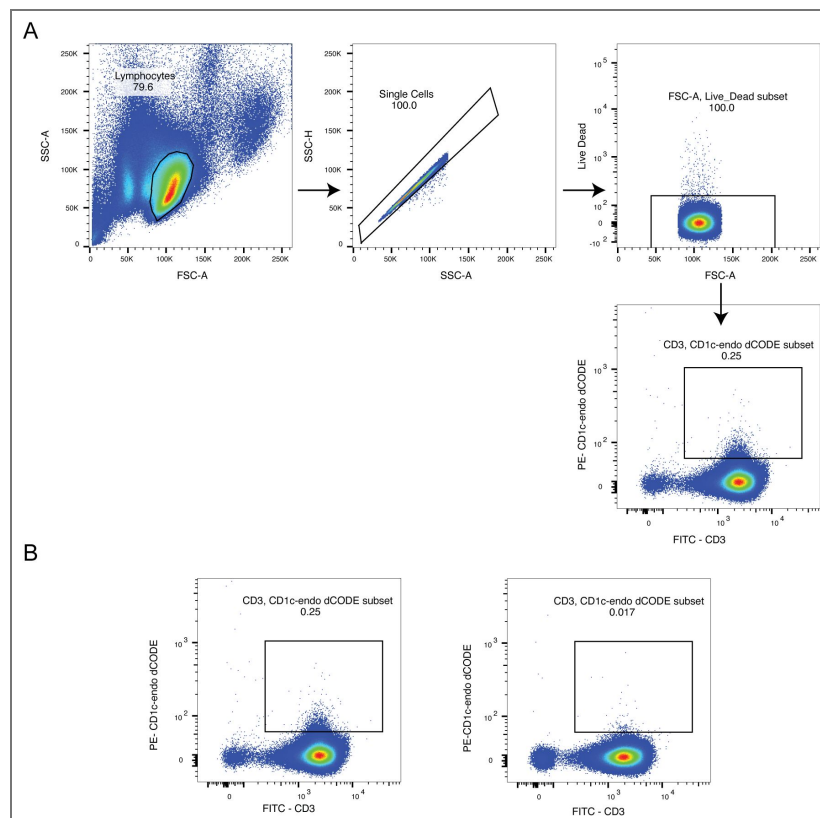


Figure S8. Dot plot showing the top 5 ranked marker genes for each annotated T-cell population, derived from all high-quality T-cells (n = 11,804).

Dot size represents the proportion of cells expressing each gene within the corresponding cluster, while colour indicates the scaled average expression (z-score). This visualization highlights the transcriptional signatures that define distinct T-cell subsets, including naïve, memory, effector, and metabolically active populations.

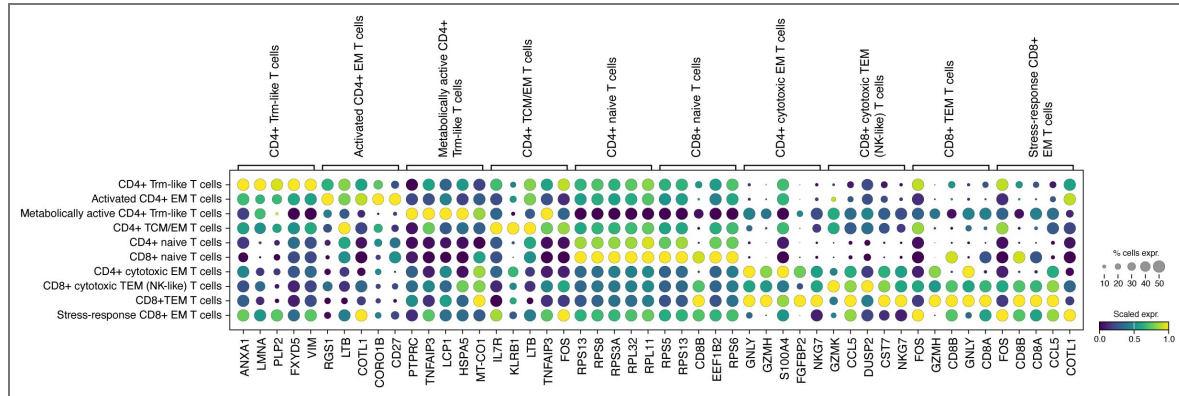
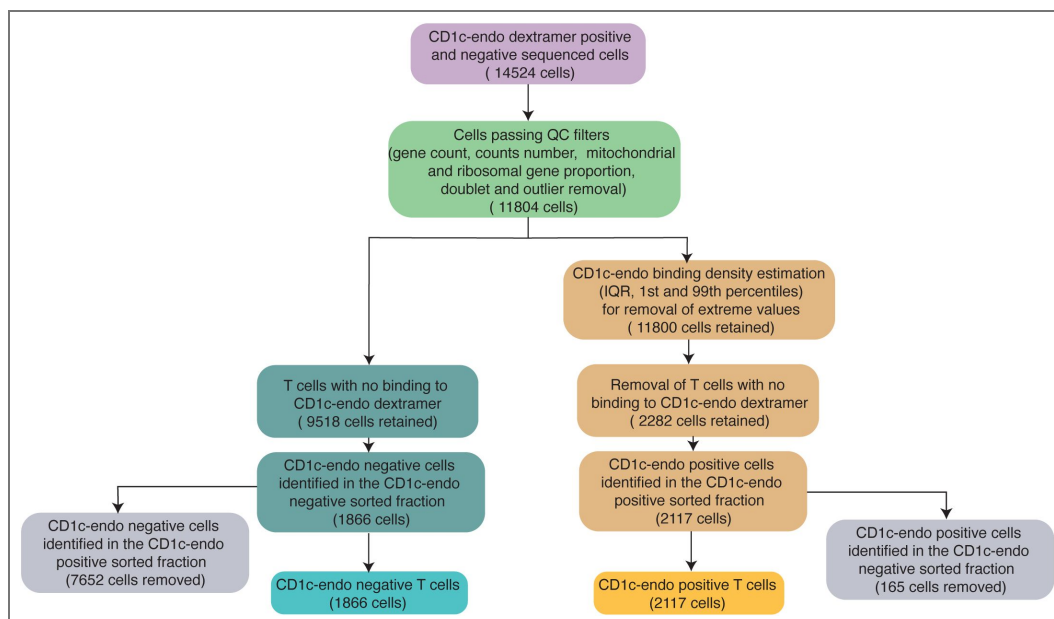


Figure S9. Filtering strategy for identifying CD1c-endo dextramer-bound T-cells in scRNA-seq data from sorted populations.

A total of 14,524 cells were initially profiled by scRNA-seq from dextramer-sorted fractions. Following standard quality control, including thresholds for gene and UMI counts, mitochondrial and ribosomal content, doublet detection, and outlier removal, 11,804 high-quality cells were retained. CD1c-endo-negative cells (n = 9,518) were defined by the absence of detectable dextramer signal and further stratified by their sort gate: 7,652 originated from the CD1c-endo-positive gate and 1,866 from the CD1c-endo-negative gate. To minimise false negatives, only the 1,866 cells from the negative sort gate were retained as confidently CD1c-endo-negative. To define CD1c-endo-positive cells, dextramer signal intensities were modelled using kernel density estimation (KDE) and quantile-based filtering (1st and 99th percentiles, IQR). Cells with signal below the lower bound ($Q1 - 1.5 \times IQR$) or extreme values were excluded, yielding 11,800 cells. Of these, 2,288 cells with no binding signal were removed. Among the remaining 2,282 CD1c-endo-positive candidates, 2,117 were from the CD1c-endo-positive sort gate and 165 from the negative gate. To ensure specificity, only the 2,117 cells from the positive sort gate were retained as confidently CD1c-endo-positive. This combined quality control and gating strategy enabled robust identification of CD1c-endo-binding T-cells in the single-cell dataset.



Data availability

scRNA-seq data have been deposited in the European Nucleotide Archive (ENA) under accession number PRJEB94332 and are publicly available as of the date of publication. Additional data and analysis code are available from the corresponding author upon request.

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Additional information

Author contributions

Joint first author order reflects contribution to data generation. M.M., S.H.F., D.G.B. and S.M. designed research; M.M., S.H.F., J.G., D.G.B., D.B., S.S., R.S-K., P.T-S., A.L., R.S., A.V., L.T., A.W., L.D., P.E., and S.M. performed research; R.S-K., A.L., M.L., D.K.C., S.S., A.L., A.V., and P.E. contributed new reagents/analytic tools; M.M., S.H.F., J.G., D.G.B., L.T., A.W., M.L., L.D., D.K.C., A.L., S.S., L.T., A.L., A.V., P.E., and S.M. analyzed data; and S.M. wrote the paper, all authors edited and approved final version.

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Peer reviews

Reviewer #1 (Public review):

Summary:

T cells that recognize lipids - CD1c - are frequent in circulation; however, their role in infection is unclear. This study aims to understand how Mtb infection can shape the responses of CD1c-specific T cells. CD1c is expressed in MTB granuloma, but in lower amounts than in nearby inflamed tissue. Mtb infection downregulates the expression of CD1c on monocyte-derived DCs. Single-cell RNA sequencing revealed the cytotoxic program inherent to the lipid-CD1c-specific T cells. Using an in vitro APC system where CD1c expression remains intact upon Mtb infection, the authors suggest that these T cells react better to Mtb-infected than uninfected Cd1c-expressing APC and reduce Mtb burden in infected cells. Therefore, Cd1c downregulation could be an immune evasion strategy used by Mtb.

Strengths:

This study asks an important question. The single-cell transcription analysis suggests the inherent cytotoxic program of lipid-CD1c cells and provides insights into their phenotypic and potential functional profiles. Function experiments suggest that these autoreactive T cells can react to Mtb infection, adding to the paradigm of infection control by these non-conventional T cell populations.

Weaknesses:

The study lacks sufficient rigor; conclusions may be strengthened with the incorporation of more controls, and some deeper characterization of the THP1 system and the CD1c-specific T cells isolated from blood. Crucial conclusions are drawn from the cell mixing experiments involving the engineered THP-1 system and CD1c-lipid-specific T cells from blood. These cells need more in-depth characterization. The expression of MHC-I/II is clearly reduced in THP1-CD1c cells. However, it is important to ensure that it is completely abolished, since a residual expression can skew the result with activation of conventional T cells in the blood or low levels of conventional T cells that may be present in the CD1c-tetra/multimer sorted T cells. CD1c-tetra/multimer sorting should include more markers than used in this study.

Figure 2: The immunohistochemistry appears to be shown only for one biopsy; it may be worth quantifying the immunohistochemistry of all five. The expression of CD1 molecules goes up during the differentiation of MoDC. And Mtb infection prevents or dampens the upregulation. Does Mtb infection downregulate the CD1 expression of mature DCs? Can the effect of Mtb on the expression of CD1a,b,c molecules be investigated using CD1c-expressing DCs from blood? What could be the reason THP-1 cells do not downregulate CD1 molecules upon Mtb infection, and how about the expression of CD1a and b?

Figure 3: (F) What does the X-axis read for the no infection group? The value for MOI = 0 should be incorporated for the infected T cell group.

Figure 4: In the lysis assay, THP1-CD1c cells (uninfected and infected) incubated alone should be incorporated.

Figure 5: A quantitative brief on the single cell TCR sequencing - including how many T cells were sequenced and the frequency of different clone including EM1 and EM2 - should be shown.

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Reviewer #2 (Public review):

Summary:

The study by Milton et al titled "Human CD1c-autoreactive T cells recognise Mycobacterium tuberculosis-infected antigen-presenting cells and display cytotoxic effector programmes" characterises CD1c-restricted autoreactive T cells and their potential role in controlling Mtb infection. The authors develop a well-controlled system to assay for the functioning/activation of autoreactive T cells. They report the presence of CD1c-restricted autoreactive T cells in the circulating blood of healthy donors. They show that these T cells respond to CD1c and get activated even in the absence of any exogenous antigen. They next show that CD1c, along with CD1a and b, are typically downregulated on APCs during Mtb infection. These autoreactive T cells are cytotoxic, indicating they respond to Mtb treatment and/or to changes in the T cell ratio. The autoreactive T cells could effectively lyse Mtb-infected or PAMP-stimulated CD1c+APCs. Next, using TCR sequencing, they show that T cell responses were mediated by specific TCR clones with common sequence features. They show that these autoreactive T cells could curtail Mtb growth as measured by luminescence. Finally, using scRNAseq, they selectively identify the CD1c-reactive T cell pool and detect enrichment of typical effector memory CD4 and CD8 cells expressing cytolytic markers such as Granzyme, granulysin, etc. The lung biopsy staining, along with the other data presented here, suggests that while CD1c-restricted T cells could have potential anti-bacterial roles, Mtb downregulation effectively shuts down this mechanism for TB control.

Strengths:

The study is designed well and has developed many exciting tools to generate specific information.

Weaknesses:

The study has weaknesses in two important parameters - novelty and relevance in controlling TB. Further, the results could be better presented and discussed to allow easy understanding of the experimental design. For example, at several places, UV-killed or live Mtb were used. What is the rationale behind that? Why use irradiated THP1-CD1c cells for activating T cells?

While functional assays identified only CD4+ cells as CD1c-restricted, scRNAseq shows that both CD4+ and CD8+ cells exhibit this phenotype. Identifying the specific lipid antigen presented by CD1c could add greater value to the study.

Since autoreactivity was independent of exogenous antigen, the cytotoxic activity should also be independent of exogeneous antigens? What additional signal a THP1-CD1c cells treated with UV-killed Mtb express that is absent from the untreated cells?

The relative Mtb growth assay is confusing. CD1c cells with Mtb infection triggers massive lytic response, as shown in Figure 4. Under similar conditions, in Figure 6, the authors report a significant decline in Mtb growth in these cells. The problem is that with the kind of lytic response observed, a lot more Mtb could be present extracellularly and would evade killing. How do we reconcile the two observations?

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Reviewer #3 (Public review):

Summary:

Despite the rising global prevalence of TB, the role of non-classical T-cell pathways in host immunity remains unclear. The present study by Milton et al. is a novel contribution to the field of unconventional T-cell immunity in Mtb infection. The study addresses the role of CD1c-autoreactive T-cells and demonstrates that upon Mtb infection, these cells are significantly activated, resulting in increased cytokine production and cytotoxicity, and a reduction in the bacterial burden, specifically against Mtb-infected CD1c⁺ APCs (antigen-presenting cells). This defines their role as a plausible candidate for lipid-directed immunity in TB, complementary to MHC-restricted responses.

Strengths:

The manuscript is well written, and the novelty, impact, and limitations of this study are precisely highlighted by the authors.

Weaknesses:

The authors mention that they did not identify any specific lipids presented by CD1c on Mtb-infected APCs, making it unclear whether they are of host or bacterial origin. This leaves a gap in understanding why the response is enhanced in Mtb-infected cells, whether it is through altered self-presentation of lipids arising from Mtb-induced changes, infection-induced stress signals, or Mtb lipids, or through CD1c-dependent co-stimulation/infection signals. Direct lipid identification via lipidomics/MS of CD1c-bound lipids from Mtb-infected APCs would clarify whether the enhancement arises from altered self-lipids or subtle Mtb lipids.

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